ASSET MANAGEMENT SYSTEM

AN INDIVIDUAL PROJECT REPORT

Submitted by

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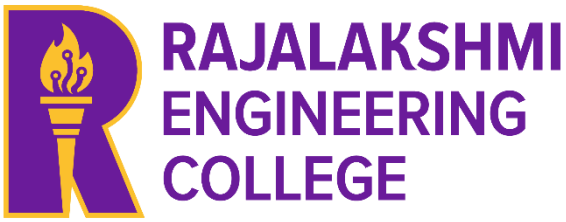
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OCB2331 – NO-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF ENGINEERING

MECHANICAL ENGINEERING



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(An Autonomous Institution, Affiliated to Anna University, Chennai)

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BONAFIDE CERTIFICATE

Certified that this project report titled "Asset Management System" is the bonafide work of AVINAASH M (241101014), Mechanical Engineering, who carried out the individual project work on the Zoho Creator no-code platform for the Open Elective course OCB2331 – No-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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INTERNAL EXAMINER

EXTERNAL EXAMINER

DECLARATION

I, AVINAASH M (241101014), hereby declare that the project report titled “Asset Management System”, submitted for the End Semester Examination of the Open Elective course OCB2331 – No-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its forms, reports, pages, rules and workflows — was designed and configured by me using the no-code features of the platform.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder) or external resource used in this work has been duly acknowledged in the report.

Place: Chennai
Date: 31 October 2026

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AVINAASH M

ABSTRACT

Mechanical engineering workshops and laboratories depend on machines, testing equipment and instruments worth several lakhs of rupees, yet the records of these assets are still kept in handwritten stock registers, issue slips and scattered spreadsheets. As a result, purchase requests wait for signatures, the location and condition of an asset are uncertain, warranty dates are missed, preventive maintenance is forgotten and the cost of repairs is never summarised for the Head of the Department.

This project presents the Asset Management System, a complete business application built on Zoho Creator without writing code. A no-code platform suits the problem because a mechanical engineer can design forms, rules and automation visually and obtain hosting, security and a mobile app without a software team. The application contains eight forms (Asset, Asset Category, Vendor, Asset Request, Asset Allocation, Maintenance Schedule, Maintenance Record with a spare-parts subform, and Asset Disposal), nine reports, a dashboard page and a quick-action page. Eight workflows were configured with the visual builder: three form workflows that keep records and statuses consistent, three approval processes for purchase, repair-cost and disposal decisions, and two schedules that send maintenance and warranty reminders. Four roles — Administrator, Supervisor, Technician and Requester — see only the data meant for them.

The application was checked with twenty test cases and three acceptance users, published to the web and the Zoho Creator mobile app on 20 October 2026, and demonstrated by following a purchase request from approval through allocation to a repair job. In the trial, the time to complete one purchase request fell from about five working days to under one day, handwritten registers were replaced by a single searchable record, and the summary report for the Head of the Department, which earlier took about two hours to prepare, is now available instantly on the dashboard.

Keywords: *No-Code Development; Zoho Creator; Citizen Developer; Workflow Automation; Dashboard; Asset Management*

PROJECT AT A GLANCE

Attribute	Details
Project Title	Asset Management System
Engineering Domain	Mechanical – Workshop and Laboratory Asset Management (procurement, allocation, preventive maintenance and disposal)
Student Name & Register No.	AVINAASH M – 241101014
Branch & Semester	Mechanical Engineering – Semester V
Zoho Creator Edition	Education
Application Name	Asset Management System
Published / Live URL	https://creatorapp.zoho.in/avinaashm/asset-management-system/
Demo Credentials (per role)	Administrator: admin.demo@ams-demo.example | Supervisor: hod.demo@ams-demo.example | Technician: tech.demo@ams-demo.example | Requester: staff.demo@ams-demo.example (common demo password: Ams@2026)
Built from	Scratch, with AI app builder (Zia) used for the first draft and Excel import for master data
Number of Forms / Subforms	8 / 1
Number of Reports (by type)	List: 4 | Kanban: 1 | Calendar: 1 | Pivot/Chart: 2 | Others: 1 (Spreadsheet)
Number of Pages / Dashboards	2 (Asset Dashboard, Quick Actions)
Number of Workflows	Form: 3 | Schedule: 2 | Approval: 3 | Blueprint: 0
Rules & Formula Fields	18 (8 formula fields, 6 field rules, 4 validation rules)
Roles Defined	Administrator, Supervisor, Technician, Requester
Mobile App / Portal Used	Yes (Zoho Creator mobile app); portal not used
Date of Publishing	20 October 2026

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
AI	Artificial Intelligence
AMS	Asset Management System
CMMS	Computerised Maintenance Management System
CNC	Computer Numerical Control
CRUD	Create, Read, Update, Delete
CSV	Comma-Separated Values
FR	Functional Requirement
HoD	Head of the Department
KPI	Key Performance Indicator
NCDP	No-Code Development Platform
NFR	Non-Functional Requirement
PM	Preventive Maintenance
QR	Quick Response (code)
RBAC	Role-Based Access Control
SDLC	Software Development Life Cycle
TC	Test Case
UAT	User Acceptance Testing
UC	Use Case
UI	User Interface
UTM	Universal Testing Machine
Zia	Zoho’s AI assistant

CHAPTER 1
INTRODUCTION

1.1 Background

A mechanical engineering department runs on its machines. The workshops and laboratories of a typical department — the Machine Shop, CAD-CAM Lab, Materials Testing Lab, Thermal Lab, Welding Shop and Metrology Lab — hold lathes, milling machines, a universal testing machine, welding sets, compressors, 3D printers, precision instruments and computers. In this project the case study is a workshop and laboratory complex with about 150 such assets worth roughly ₹85 lakh. The assets are used by students in practical classes, by faculty for research and consultancy, and by technicians who keep them running.

Asset management is the discipline of getting the best value from physical assets throughout their life, from the decision to buy, through use and maintenance, to retirement [1]. For a mechanical department this life cycle has five stages: a staff member asks for a new item, the request is approved and the item is purchased, the item is registered and tagged, it is allocated to a laboratory or person and maintained, and finally it is disposed of when it is obsolete or beyond repair. Preventive maintenance, which means servicing a machine on a fixed schedule before it fails, is known to reduce unplanned downtime and repair cost [2].

Today these stages are handled on paper. The Lab in-charge writes a requisition, the Head of the Department signs it, the store keeper enters the item in a handwritten stock register, and a clerk copies the details into a spreadsheet when time permits. Breakdowns are reported verbally and repair bills are filed in a folder. The people involved are the Head of the Department (HoD), the asset manager in charge of the stores, lab in-charges and faculty who request items, and technicians who maintain the machines.

1.2 Problem Statement

The department has no single, reliable record of its assets and no automatic way of moving a request, a repair or a disposal through the people who must approve it. Information is spread over registers, slips and spreadsheets, so the staff cannot quickly answer simple questions such as where a machine is, whether it is still under warranty,

when it was last serviced, how much it has cost to repair, or which purchase requests are waiting for approval.

- **Lost and duplicate records:** handwritten registers and issue slips are lost, damaged or entered twice, so the stock position is never certain.
- **Delayed approvals:** purchase and disposal files move by hand from desk to desk, and nobody can see where a file is waiting.
- **No maintenance reminders:** preventive servicing depends on memory, so machines are serviced late or not at all and fail during practical classes.
- **Missed warranty claims:** warranty expiry dates are not tracked, so repairs that the supplier should have paid for are paid by the department.
- **No summary for management:** the cost history of each machine and the department-wide asset position take hours to prepare for the HoD.

1.3 Objectives

The objectives of the project are listed below. Each objective is implemented in a module of Chapter 5 and its result is discussed in Chapter 8.

- To digitise asset, category, vendor, request, allocation, maintenance and disposal records using eight Zoho Creator forms with validations, lookups and a spare-parts subform.
- To provide nine reports (list, Kanban, calendar, chart, pivot and spreadsheet) and a dashboard page for monitoring asset status, value, maintenance cost and pending approvals.
- To automate notifications, status updates and reminders using three form workflows and two schedules built with the visual workflow builder.
- To route purchase requests, high-value repairs and disposals through three approval processes with levels, conditions and e-mail notifications.
- To restrict access to data and screens according to four user roles and record-level rules.
- To publish the application for web and mobile use by the department’s staff and technicians.

1.4 Scope of the Project

In scope. The application covers the complete life cycle of workshop and laboratory assets: master data (categories and vendors), asset registration with automatic depreciation and warranty status, purchase requests with approval, allocation and return of assets, preventive maintenance schedules with reminders, maintenance jobs with spare parts and cost, repair-cost approval, and disposal with approval. It provides reports and a dashboard for the Administrator and the Supervisor, and role-based access for four types of users: Administrator (asset manager), Supervisor (HoD), Technician and Requester (faculty and lab in-charges).

Out of scope. Accounting entries, payroll, inventory of consumables and raw materials, automatic reading of machine sensors, and purchase-order generation with external suppliers are not included. The application is accessed through a web browser and the Zoho Creator mobile app; a customer portal for outside users is not used.

1.5 Why No-Code?

A no-code development platform lets a person who is not a software engineer build a working business application using drag-and-drop builders, properties, rules and a visual workflow editor instead of programming. Zoho Creator provides these tools together with hosting, a database, user management, security and a mobile app [3]. Such platforms are part of the wider low-code movement, in which people close to the business problem — often called citizen developers — build their own applications [4]. Studies of manufacturing companies report that low-code platforms shorten development time and help engineers digitise their own processes without waiting for a software team [5], and market analysts consider the category a mainstream way of delivering enterprise applications [6].

For this project a no-code platform has five clear advantages. First, the application could be built and changed quickly by the author, a mechanical engineering student, because the process knowledge is in the author's own discipline. Second, no server, database or mobile app had to be installed or programmed. Third, approvals, reminders and status updates are available as ready-made workflow blocks. Fourth, roles and record-level access are configured rather than coded. Fifth, new fields, reports or rules can be added in minutes when the department's needs change.

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related literature on no-code platforms and asset management software. Chapter 3 presents the requirement analysis. Chapter 4 describes the application design. Chapter 5 details how the application was built on Zoho Creator. Chapter 6 covers testing. Chapter 7 discusses publishing, sharing and security. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements. The appendices contain the feature checklist, course outcome mapping, configuration record, user manual, evaluation rubric and viva-voce record sheet.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

The existing system was studied by following one asset through its life in the Mechanical Engineering workshops. The steps, shown in Figure 2.1, are as follows. (1) A staff member identifies the need for a new item or the failure of a machine. (2) A paper requisition is written and signed by the lab in-charge. (3) The file is carried to the HoD for approval and returned to the store. (4) The store purchases the item and files the bill. (5) The store keeper enters the item in the handwritten stock register, and a clerk later copies it into an Excel sheet. (6) When the item is given to a laboratory, an issue slip is filled. (7) When a machine breaks down, the fault is reported verbally to a technician. (8) The repair is done without a job card, and the bill is kept in a folder.

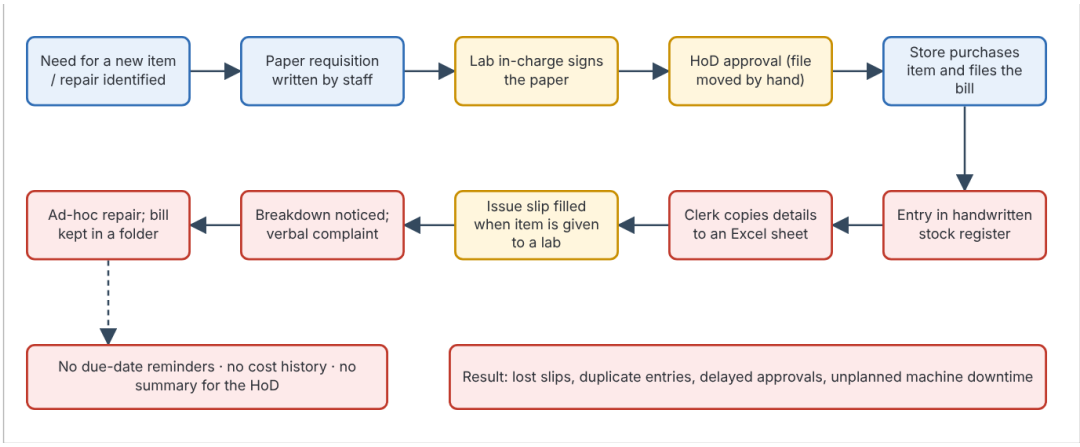


Figure 2.1: As-Is Process Flow of the Existing System

2.2 Limitations of the Existing System

- Stock registers and issue slips are paper documents that can be lost, damaged or filled incorrectly; the Excel copy is always behind the register.
- Approvals depend on a physical file, so no one knows at which desk a request is waiting and requests are delayed by days.
- There is no reminder for preventive maintenance, calibration or warranty expiry; the dates exist only in the memory of individual technicians.
- The history of repairs and spare parts used for each machine is not recorded, so the real cost of owning a machine cannot be calculated.
- The location and status of an asset (in stock, in use, under maintenance or disposed) cannot be confirmed without walking to the laboratory.

- The HoD has no dashboard; reports for audits and accreditation must be prepared manually and take several hours.

2.3 No-Code, Low-Code and Traditional Development

Three ways of building an asset management application were compared for this project. Table 2.1 summarises the comparison for the proposed application.

Table 2.1: Comparison of Development Approaches for the Proposed Application

Criterion	Traditional (Coding)	Low-Code	No-Code (This Project)
Programming skill needed	High: web, database and mobile programming	Medium: scripting for most logic	None: drag-and-drop builders and rules
Development time	Several months	Several weeks	About three weeks, part-time
Hosting & mobile app	Server and separate app to be built and maintained	Platform hosting; mobile app partly configured	Built-in hosting and ready mobile app
Ease of making changes	Needs a developer and a new release	Moderate; scripts to be edited	Edit the form or rule and save
Flexibility / customisation	Unlimited	High	Moderate; limited to platform features
Cost	High (developers, servers)	Medium (licences and developers)	Low (education edition, no developer)

2.4 Literature Survey

Twelve sources were reviewed: standards and books on asset and maintenance management, research papers on low-code and no-code development, the Zoho Creator documentation and learning material, and commercial or open-source asset management tools. The most relevant ones are summarised in Table 2.2.

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	ISO, 2014 [1]	ISO 55000: Asset management	Defines asset management as realising value from assets across their whole life cycle.	Basis for the five life-cycle stages modelled in the application.
2	R. K. Mobley, 2002 [2]	An Introduction to Predictive Maintenance	Planned maintenance and good records reduce downtime and repair cost.	Justifies the Maintenance Schedule form and reminder workflow.
3	Bock and Frank, 2021 [4]	Low-Code Platform	Describes low-code platforms and citizen developers as a way to speed up application delivery.	Supports the choice of a no-code platform built by a non-programmer.

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
4	Sanchis et al., 2020 [5]	Low-code as enabler of digital transformation in manufacturing industry	Low-code helps manufacturing engineers digitise their own processes quickly.	Shows relevance to mechanical and manufacturing contexts.
5	Zoho Corporation [3]	Zoho Creator Help Documentation	Explains forms, field types, reports, pages, workflows, approvals and roles.	Primary reference for every feature configured.
6	A. Shabdar, 2017 [7]	Mastering Zoho Creator	Step-by-step guide to building business applications on Zoho Creator.	Guided form design, lookups and report planning.
7	UpKeep [8]	UpKeep maintenance and asset management	Mobile-first work orders, preventive maintenance and asset records.	Confirms features expected in modern CMMS tools; paid and generic.
8	Fiix [9]	Fiix CMMS	Cloud CMMS with work orders, preventive schedules and asset history.	Benchmark for maintenance workflow; subscription cost is high for a department.
9	Snipe-IT [10]	Snipe-IT open-source asset management	Open-source tool for tracking assets, check-out and check-in.	Shows allocation and return logic; needs server setup and has no approval flow.

2.5 Proposed System

The proposed system is the Asset Management System, a Zoho Creator application that replaces registers, slips and spreadsheets with eight linked forms and removes each limitation listed in Section 2.2. Records are entered once, with validations and lookups, and are available to every authorised user on the web and on mobile. Approval workflows carry purchase, repair-cost and disposal requests to the right person automatically and tell the requester the result. Schedules send maintenance and warranty reminders each day or week. Formula fields calculate age, book value, warranty status and repair cost, so no manual calculation is needed. Nine reports and a dashboard give the HoD the asset position, cost history and pending approvals instantly. Role-based access shows each user only the records meant for them.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Users and Roles

Four roles were identified after discussions with the workshop staff. Each role is created in Zoho Creator and given a permission set (Section 5.7).

Table 3.1: Users and Roles

Role	Who (in the organisation)	What the Role Does in the Application
Administrator	Asset manager / stores in-charge	Manages master data, registers assets, allocates and receives assets, plans maintenance, gives first-level approval, and views all reports and the dashboard.
Supervisor	Head of the Department	Gives final approval for purchases above ₹25,000, repair costs above ₹10,000 and disposals; views the dashboard and all reports.
Technician	Workshop and laboratory technicians	Views assigned maintenance schedules, creates and updates maintenance records with spare parts and labour cost.
Requester	Faculty and lab in-charges	Raises purchase and disposal requests and tracks the status of their own requests.

3.2 Functional Requirements

The functional requirements are listed in Table 3.2. Each requirement is testable and is referred to by its FR ID in the test cases of Chapter 6.

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The application shall allow the Administrator to register an asset with mandatory fields (name, category, serial number, vendor, location, purchase date, cost and warranty expiry).	Administrator	High
FR-02	The application shall calculate asset age, current book value and warranty status automatically.	System	High
FR-03	The application shall maintain master lists of asset categories and vendors, with unique names.	Administrator	Medium
FR-04	The application shall allow a requester to raise a purchase request and calculate the estimated total cost automatically.	Requester	High
FR-05	The application shall send a purchase request for approval to the Administrator and, when the cost exceeds ₹25,000, to the Supervisor, and inform the requester of the result.	Administrator, Supervisor	High

FR ID	Requirement	Role(s)	Priority
FR-06	The application shall allocate an asset to a person or laboratory, change the asset status to In Use, and restore it to In Stock when the asset is returned.	Administrator	High
FR-07	The application shall store a preventive maintenance schedule for an asset and calculate the next due date from the frequency.	Administrator	High
FR-08	The application shall remind the assigned technician three days before the due date and inform the Supervisor if the maintenance is overdue.	Technician, Supervisor	High
FR-09	The application shall allow a technician to record a maintenance job with spare parts (subform), labour cost and automatic total cost, and shall ask for the fault description only for breakdown jobs.	Technician	High
FR-10	The application shall send a completed job with total cost above ₹10,000 to the Supervisor for approval.	Supervisor	Medium
FR-11	The application shall update the schedule's last-done date and the asset status when a job is completed.	System	High
FR-12	The application shall allow a disposal request to be raised and approved at two levels, and set the asset status to Disposed on approval.	Administrator, Supervisor	Medium
FR-13	The application shall e-mail the Administrator every Monday a list of assets whose warranty expires within 30 days.	Administrator	Medium
FR-14	The application shall show a dashboard with KPIs and charts, and reports in list, Kanban, calendar, chart, pivot and spreadsheet forms.	Administrator, Supervisor	Medium
FR-15	The application shall restrict each role to the forms, reports and records permitted to it, including record-level rules for technicians and requesters.	All roles	High
FR-16	The application shall be usable on the Zoho Creator mobile app, including photo capture and barcode / QR scanning for assets.	Administrator, Technician	Medium

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Usability	A new user shall be able to submit a request or a job record without training, using clear labels and default values.
NFR-02	Security	Users shall see only the data allowed for their role; sensitive fields such as cost shall be hidden from requesters.
NFR-03	Accessibility	The application shall work on web browsers and on the Zoho Creator mobile app (Android and iOS).
NFR-04	Reliability	Records shall be stored in the Zoho cloud with automatic backup; no record shall be lost when a user's device fails.

NFR ID	Category	Requirement
NFR-05	Performance	Forms and reports shall open in under three seconds for up to 5,000 records on a normal broadband connection.
NFR-06	Data integrity	Mandatory fields, unique serial numbers and validation rules shall prevent incomplete or duplicate records.
NFR-07	Maintainability	A form, rule or report shall be changeable by the administrator without programming or downtime.
NFR-08	Auditability	Every record shall show who added or modified it and when, through the record history.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator Education Edition (cloud)
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox (latest version)
Mobile	Zoho Creator mobile app (Android / iOS) with camera access for photo and barcode / QR scanning
Other Zoho services	Zoho Mail (notification e-mails) and Zoho Sheet (report export), used through built-in features
Client hardware	Any internet-enabled desktop, laptop or smartphone; no installation required

3.5 Use Case Model

The use case diagram in Figure 3.1 shows the four actors and the main functions of the system. Three key use cases are described in Tables 3.5 to 3.7.

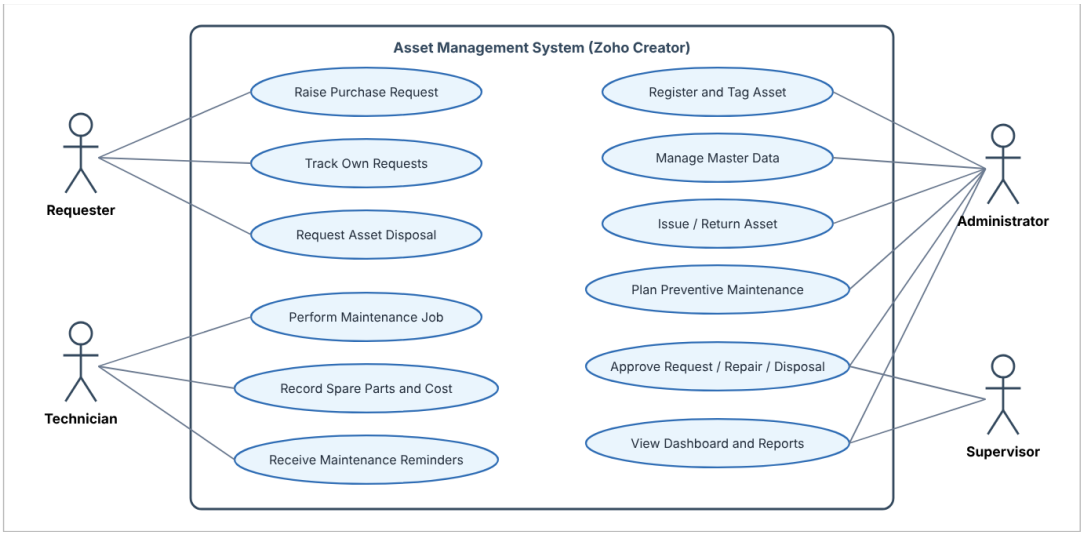


Figure 3.1: Use Case Diagram of the Application

Table 3.5: Use Case Description – Raise and Approve a Purchase Request

Use Case ID	UC-01
Use Case Name	Raise and approve a purchase request
Actor(s)	Requester, Administrator, Supervisor
Pre-condition	The requester has logged in; the item's category exists in the Asset Category form.
Main Flow	1. The requester opens the Asset Request form and fills the item, quantity, unit cost, priority and justification. 2. The estimated total cost is calculated automatically. 3. On submission, workflow W4 sends the request to the Administrator for Level 1 approval. 4. If the total cost is above ₹25,000, the request goes to the Supervisor for Level 2 approval. 5. The approval status is updated and the requester receives an e-mail.
Alternate Flow	An approver rejects the request with comments; the status becomes Rejected, the rejection reason is shown and the requester is informed.
Post-condition	The request shows Approved or Rejected and is visible in the Purchase Requests report.
Related FR	FR-04, FR-05, FR-15

Table 3.6: Use Case Description – Perform Preventive Maintenance

Use Case ID	UC-02
Use Case Name	Perform preventive maintenance and record cost
Actor(s)	Technician, Supervisor, Administrator
Pre-condition	A maintenance schedule exists for the asset and a technician is assigned.
Main Flow	1. The daily schedule W7 sends a reminder to the technician three days before the due date. 2. The technician opens the Maintenance Record form, selects the asset and schedule and enters the work done. 3. Spare parts and labour cost are entered; parts cost and total cost are calculated automatically. 4. The technician sets the job status to Completed. 5. Workflow W3 updates the schedule's last-done date, recalculates the next due date and restores the asset status.
Alternate Flow	If the total cost exceeds ₹10,000, workflow W5 sends the job to the Supervisor; a rejection returns the job to In Progress.
Post-condition	The asset shows the updated status and the job appears in the Completed column of the Job Board.
Related FR	FR-07, FR-08, FR-09, FR-10, FR-11

Table 3.7: Use Case Description – Allocate an Asset to a Laboratory

Use Case ID	UC-03
Use Case Name	Allocate and receive back an asset

Actor(s)	Administrator
Pre-condition	The asset is registered and its status is In Stock.
Main Flow	1. The Administrator opens the Asset Allocation form and selects an In Stock asset from the filtered lookup. 2. The person, laboratory, issue date and expected return date are entered and the receiver signs on the Signature field. 3. Workflow W2 sets the asset status to In Use. 4. On return, the Administrator enters the actual return date and the condition. 5. Workflow W2 sets the allocation to Returned and the asset to In Stock.
Alternate Flow	If the expected return date is before the issue date, validation rule VL-04 shows an error and the record is not saved.
Post-condition	The Allocation Register and the asset status reflect the issue or return.
Related FR	FR-06, FR-15

CHAPTER 4
APPLICATION DESIGN

4.1 Application Structure

Figure 4.1 shows the structure of the application. Users reach it through a web browser or the Zoho Creator mobile app. The application menu has four parts: Dashboard, Quick Actions, Forms and Reports. Underneath are eight forms, nine reports and two pages. Eight workflows connect the forms and send notifications, and role-based permissions control what each user can open. Everything runs on the Zoho Creator cloud, so no server or database has to be maintained by the department.

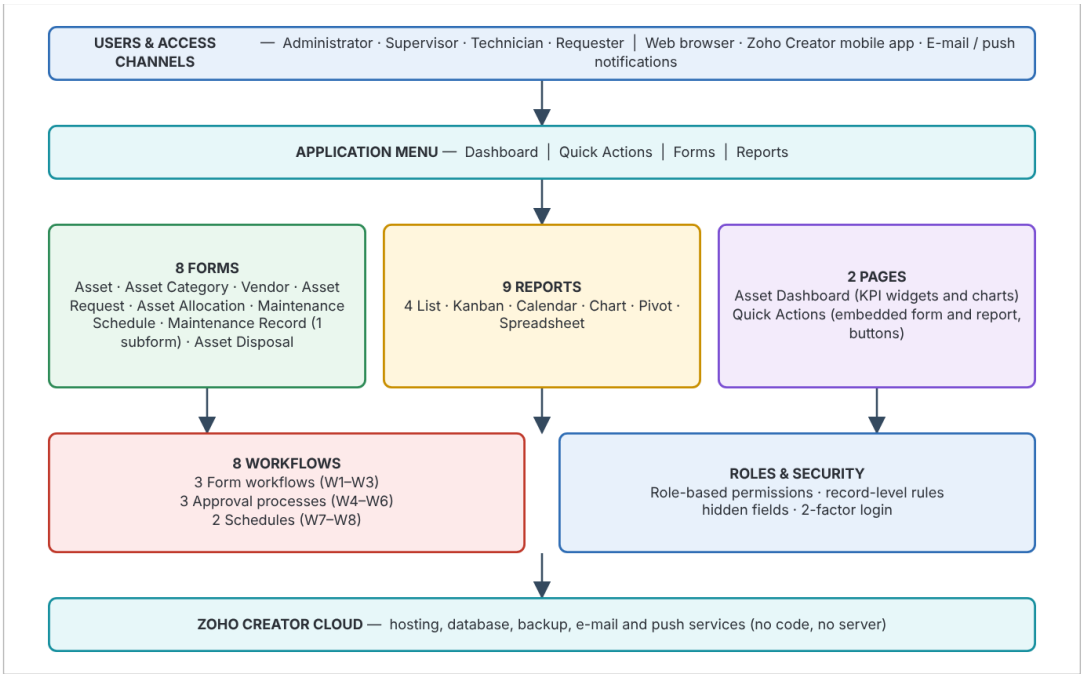


Figure 4.1: Overall Structure of the Application

4.2 Data Model

Eight forms are needed to hold the data. Asset Category and Vendor are master forms. The Asset form is the centre of the model: it looks up the category and the vendor, and the four transaction forms (Asset Allocation, Maintenance Schedule, Maintenance Record and Asset Disposal) each look up the asset. Asset Request looks up the category and an optional preferred vendor. Maintenance Record also looks up the schedule and holds a subform for spare parts. The relationships are shown in Figure 4.2 and listed in Table 4.1.

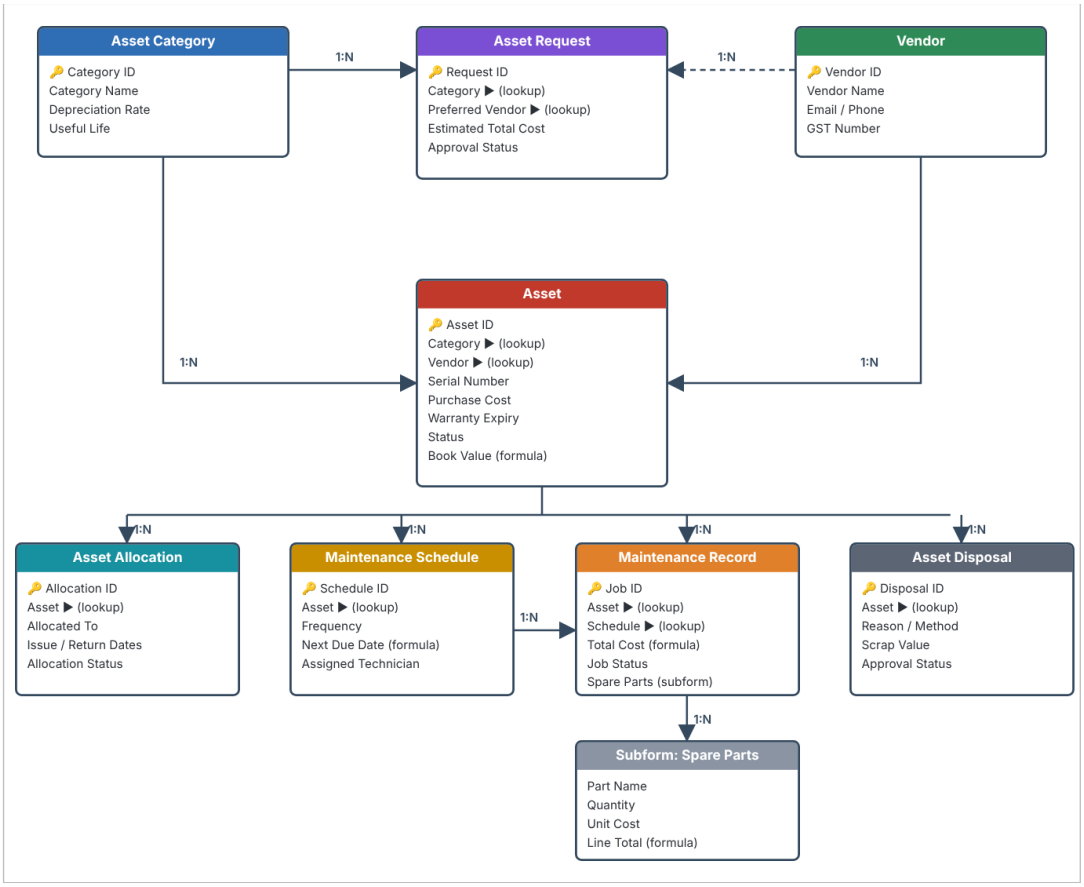


Figure 4.2: Data Model Showing Forms and Their Links

Table 4.1: Forms and Their Relationships

Form	Purpose	Linked To	Link Type	Configured Using
Asset Category	Master list of categories with depreciation rate	Asset, Asset Request	One category – many assets / requests	Lookup field
Vendor	Master list of suppliers	Asset, Asset Request	One vendor – many assets / requests	Lookup field
Asset	Central asset register	Asset Allocation, Maintenance Schedule, Maintenance Record, Asset Disposal	One asset – many records	Lookup field
Asset Request	Purchase request with approval	Asset Category, Vendor	Many requests – one category / vendor	Lookup field
Asset Allocation	Issue and return of assets	Asset	Many allocations – one asset	Lookup field (filtered to In Stock)
Maintenance Schedule	Preventive maintenance plan	Asset, Maintenance Record	One asset – many schedules; one schedule – many jobs	Lookup field

Form	Purpose	Linked To	Link Type	Configured Using
Maintenance Record	Job card with parts and cost	Asset, Maintenance Schedule, Spare Parts	One record – many parts rows	Lookup field and subform
Asset Disposal	Disposal request with approval	Asset	Many disposals – one asset	Lookup field

4.3 Form Design

A field specification is given for each form in Tables 4.2 to 4.9. Field types use the names shown on the Zoho Creator platform. In total 23 distinct field types are used: Auto Number, Single Line, Multi Line, Number, Decimal, Currency, Percent, Email, Phone, Name, Address, Date, Drop Down, Radio, Decision Box, Lookup, Users, Subform, File Upload, Image, Signature, Barcode / QR and Formula.

Table 4.2: Field Specification – Asset Category Form

Field Label	Field Type	Mandatory	Properties (Validation / Default / Unique)	Remarks
Category ID	Auto Number	—	Prefix CAT-	System generated
Category Name	Single Line	Yes	Unique	e.g., Machine Tools
Depreciation Rate (% / year)	Percent	Yes	Default 10; range 0–100	Used by Asset formulas
Useful Life (Years)	Number	Yes	Minimum 1	Default life of category
Description	Multi Line	No	—	Free-text notes
Active	Decision Box	No	Default: checked	Hides retired categories

Table 4.3: Field Specification – Vendor Form

Field Label	Field Type	Mandatory	Properties (Validation / Default / Unique)	Remarks
Vendor ID	Auto Number	—	Prefix VND-	System generated
Vendor Name	Single Line	Yes	Unique	Company name
Contact Person	Name	Yes	First and last name	—
Email	Email	Yes	Valid e-mail format	Used for purchase queries
Phone	Phone	Yes	10-digit number	—
GST Number	Single Line	No	Exactly 15 characters	Validation rule VL-02
Address	Address	No	—	City and PIN code
Vendor Rating	Radio	No	Excellent / Good / Average / Poor	Default: Good

Table 4.4: Field Specification – Asset Form

Field Label	Field Type	Mandatory	Properties (Validation / Default / Unique)	Remarks
Asset ID	Auto Number	—	Prefix AST-	System generated
Asset Name	Single Line	Yes	—	e.g., Centre Lathe
Category	Lookup	Yes	Looks up Asset Category	Sets depreciation rate
Serial Number	Single Line	Yes	Unique	Manufacturer serial no.
Vendor	Lookup	Yes	Looks up Vendor	Supplier of the asset
Location	Drop Down	Yes	Machine Shop / CAD-CAM Lab / Materials Testing Lab / Thermal Lab / Welding Shop / Metrology Lab	Physical location
Purchase Date	Date	Yes	Must not be a future date	Validation rule VL-01
Purchase Cost (₹)	Currency	Yes	Greater than 0	—
Depreciation Rate	Percent	—	Read-only; set from Category	Field rule RL-01
Warranty Expiry	Date	Yes	On or after Purchase Date	Used by schedule W8
Asset Condition	Radio	Yes	Excellent / Good / Fair / Poor	Default: Good
Status	Drop Down	Yes	In Stock / In Use / Under Maintenance / Disposed	Default: In Stock
Asset Age (Years)	Formula	—	Purchase Date to today ÷ 365	Auto-calculated
Current Book Value (₹)	Formula	—	Straight-line depreciation	Auto-calculated
Warranty Status	Formula	—	In Warranty / Expired	Auto-calculated
Asset Photo	Image	No	Capture from camera	Mobile use
Asset Tag	Barcode / QR	No	Scan on mobile app	Physical tag number
Remarks	Multi Line	No	—	—

Table 4.5: Field Specification – Asset Request Form

Field Label	Field Type	Mandatory	Properties (Validation / Default / Unique)	Remarks
Request ID	Auto Number	—	Prefix REQ-	System generated
Requested By	Users	Yes	Default: logged-in user	—
Lab / Section	Drop Down	Yes	Machine Shop / CAD-CAM Lab / Materials Testing Lab / Thermal Lab / Welding Shop / Metrology Lab	—
Category	Lookup	Yes	Looks up Asset Category	—
Item Description	Multi Line	Yes	—	Specification of item
Quantity	Number	Yes	Minimum 1	—
Estimated Unit Cost (₹)	Currency	Yes	Greater than 0	—
Estimated Total Cost (₹)	Formula	—	Quantity × Unit Cost	Auto-calculated

Field Label	Field Type	Mandatory	Properties (Validation / Default / Unique)	Remarks
Priority	Radio	Yes	Low / Medium / High	Default: Medium
Required By	Date	Yes	On or after today	Validation rule VL-03
Preferred Vendor	Lookup	No	Looks up Vendor	—
Justification	Multi Line	Yes	—	Reason for purchase
Quotation	File Upload	No	PDF, maximum 5 MB	Supporting document
Approval Status	Drop Down	—	Pending / Approved / Rejected	Updated by workflow W4
Rejection Reason	Multi Line	—	Shown only if Rejected	Field rule RL-03

Table 4.6: Field Specification – Asset Allocation Form

Field Label	Field Type	Mandatory	Properties (Validation / Default / Unique)	Remarks
Allocation ID	Auto Number	—	Prefix ALC-	System generated
Asset	Lookup	Yes	Looks up Asset; only Status = In Stock	Filtered lookup
Allocated To	Users	Yes	—	Person responsible
Lab / Section	Drop Down	Yes	Machine Shop / CAD-CAM Lab / Materials Testing Lab / Thermal Lab / Welding Shop / Metrology Lab	—
Issue Date	Date	Yes	Default: today	—
Expected Return Date	Date	Yes	On or after Issue Date	Validation rule VL-04
Actual Return Date	Date	No	Enabled on edit only	—
Allocation Status	Radio	Yes	Issued / Returned	Default: Issued
Condition on Return	Drop Down	No	Excellent / Good / Fair / Poor	Field rule RL-04
Receiver Signature	Signature	Yes	—	Acknowledgement of receipt

Table 4.7: Field Specification – Maintenance Schedule Form

Field Label	Field Type	Mandatory	Properties (Validation / Default / Unique)	Remarks
Schedule ID	Auto Number	—	Prefix MS-	System generated
Asset	Lookup	Yes	Looks up Asset	—
Maintenance Type	Drop Down	Yes	Lubrication / Calibration / Inspection / Servicing / Overhaul	—
Frequency	Drop Down	Yes	Weekly / Monthly / Quarterly / Half-Yearly / Yearly	—
Interval (Days)	Number	—	Read-only; set from Frequency	Field rule RL-02
Last Done Date	Date	Yes	Not a future date	—
Next Due Date	Formula	—	Last Done Date + Interval	Auto-calculated

Field Label	Field Type	Mandatory	Properties (Validation / Default / Unique)	Remarks
Assigned Technician	Users	Yes	—	Receives reminders
Schedule Status	Drop Down	Yes	Active / Paused / Closed	Default: Active
Instructions	Multi Line	No	—	Checklist for technician

Table 4.8: Field Specification – Maintenance Record Form

Field Label	Field Type	Mandatory	Properties (Validation / Default / Unique)	Remarks
Job ID	Auto Number	—	Prefix JOB-	System generated
Asset	Lookup	Yes	Looks up Asset	—
Schedule	Lookup	No	Looks up Maintenance Schedule	Blank for breakdown jobs
Maintenance Type	Radio	Yes	Preventive / Breakdown	Default: Preventive
Fault Description	Multi Line	Yes	Shown only for Breakdown	Field rule RL-05
Technician	Users	Yes	—	—
Job Date	Date	Yes	Default: today	—
Downtime (Hours)	Decimal	No	Two decimal places	For KPI report
Spare Parts Used	Subform	No	Part Name, Quantity, Unit Cost, Line Total	Multiple rows
Labour Cost (₹)	Currency	Yes	0 or more	—
Parts Cost (₹)	Formula	—	Sum of subform Line Totals	Auto-calculated
Total Cost (₹)	Formula	—	Labour Cost + Parts Cost	Auto-calculated
Job Status	Drop Down	Yes	Open / In Progress / Completed	Default: Open
Job Report	File Upload	No	PDF or image	Evidence of work
Approval Status	Drop Down	—	Not Required / Pending / Approved / Rejected	Updated by workflow W5

Table 4.9: Field Specification – Asset Disposal Form

Field Label	Field Type	Mandatory	Properties (Validation / Default / Unique)	Remarks
Disposal ID	Auto Number	—	Prefix DSP-	System generated
Asset	Lookup	Yes	Looks up Asset; Status ≠ Disposed	—
Reason	Drop Down	Yes	Obsolete / Beyond Repair / Replaced / Lost	—
Disposal Method	Radio	Yes	Scrap Sale / Auction / Donation / Recycle	Default: Scrap Sale
Book Value (₹)	Currency	—	Read-only; fetched from Asset	Field rule RL-06
Expected Scrap Value (₹)	Currency	Yes	0 or more	—
Disposal Date	Date	Yes	On or after today	—

Field Label	Field Type	Mandatory	Properties (Validation / Default / Unique)	Remarks
Condition Report	File Upload	No	PDF or image	Supporting document
Approval Status	Drop Down	—	Pending / Approved / Rejected	Updated by workflow W6
Remarks	Multi Line	No	—	—

4.4 Process Flow Design

Two automated processes were designed before building. Figure 4.3 shows the asset life cycle from purchase request to disposal, and Figure 4.4 shows the preventive maintenance process. Steps automated by a workflow carry a green tag with the workflow number (W1 to W8, listed in Table 5.3).

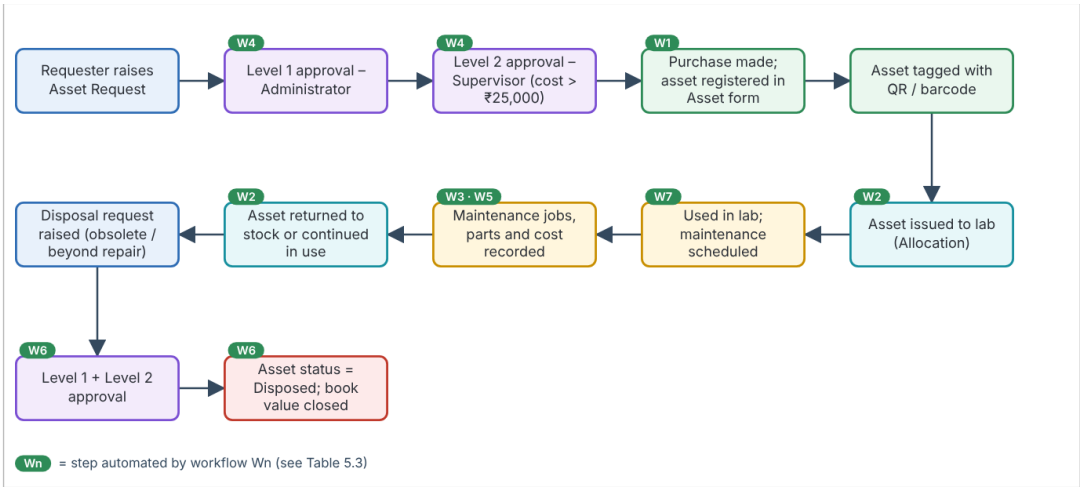


Figure 4.3: Process Flow for the Asset Life Cycle

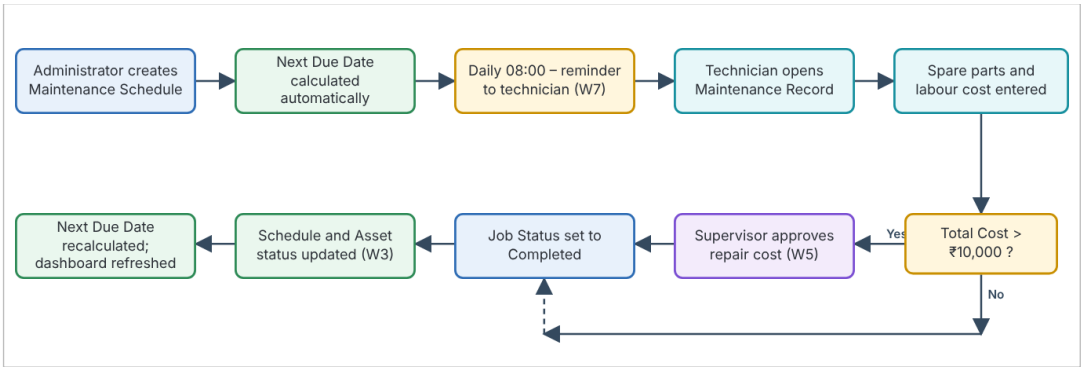


Figure 4.4: Process Flow for Preventive Maintenance

4.5 Screen and Dashboard Design

The menu was planned with the most frequent tasks first: the Dashboard and Quick Actions at the top, then the eight forms, then the nine reports. The dashboard wireframe in Figure 4.5 places six KPI tiles in a row, followed by charts for category, status and

monthly cost, and a table of upcoming maintenance, so that the HoD sees the main points without scrolling.

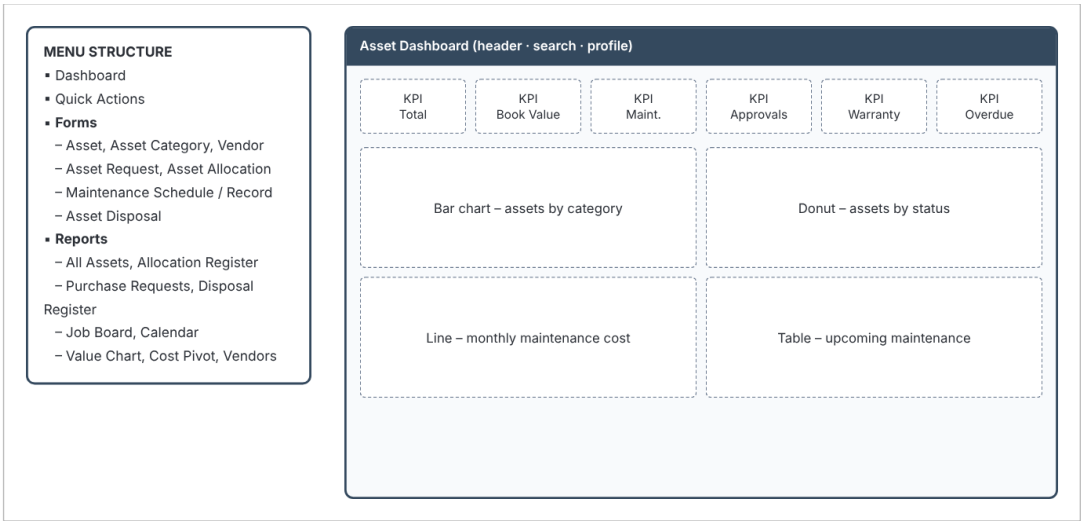


Figure 4.5: Menu Structure and Dashboard Wireframe

4.6 Access Design

Table 4.10 shows who can do what. A = Add, E = Edit, V = View, D = Delete and — = No access. Record-level rules limit technicians and requesters to their own or assigned records.

Table 4.10: Role–Permission Matrix

Component	Administrator	Supervisor	Technician	Requester
Form: Asset	A E V D	V	V	—
Form: Asset Category / Vendor	A E V D	V	—	—
Form: Asset Request	A E V D	E V	—	A V (own)
Form: Asset Allocation	A E V D	V	—	—
Form: Maintenance Schedule	A E V D	V	V (assigned)	—
Form: Maintenance Record	A E V D	E V	A E V (own)	—
Form: Asset Disposal	A E V D	E V	—	A V (own)
Report: All Assets	V	V	V	—
Report: Purchase Requests	V	V	—	V (own)
Report: Maintenance Job Board	V	V	V (assigned)	—
Page: Asset Dashboard	V	V	—	—
Page: Quick Actions	V	V	V	V

CHAPTER 5
IMPLEMENTATION ON ZOHO CREATOR

5.1 Creating the Application

The application was created in Zoho Creator under the application name “Asset Management System” (link name asset-management-system). The Zia AI app builder was used to produce a first draft from a text prompt (Section 5.9). The draft was then edited by hand: forms were renamed and extended, unnecessary forms were removed, and the Maintenance form was split into Schedule and Record. The master data of about 150 assets was imported from an Excel sheet (Section 5.8). The Zoho Creator learning material [11] was followed for the builder steps.

The menu was arranged into Dashboard, Quick Actions, Forms and Reports sections. A blue theme and an application icon were selected, and the application home page shows the six key figures and quick links to the most common tasks, as seen in Figure 5.1.

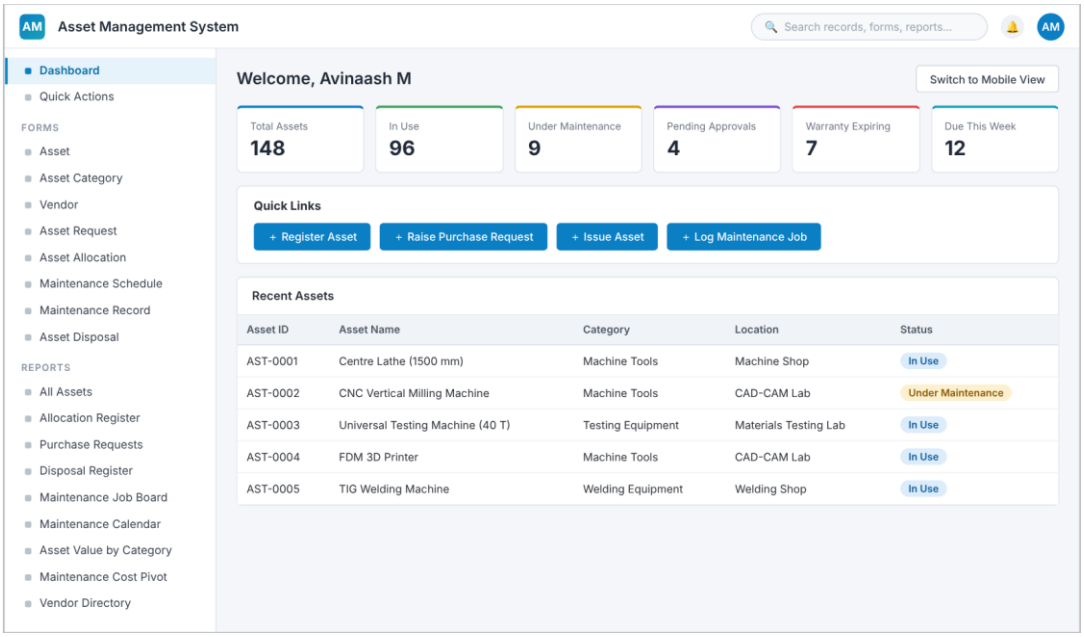


Figure 5.1: Application Home Screen and Menu

5.2 Forms

All eight forms were built with the drag-and-drop form builder by placing fields from the field-type palette on the canvas and setting properties on the right-hand panel. Mandatory, unique and default-value properties were set as listed in Chapter 4. Lookup fields link the forms, one subform holds the spare parts, and media fields (Image, File

Upload, Signature, Barcode / QR) capture evidence. Each form has a success message and the standard Submit and Reset buttons. The Asset form in the builder is shown in Figure 5.2.

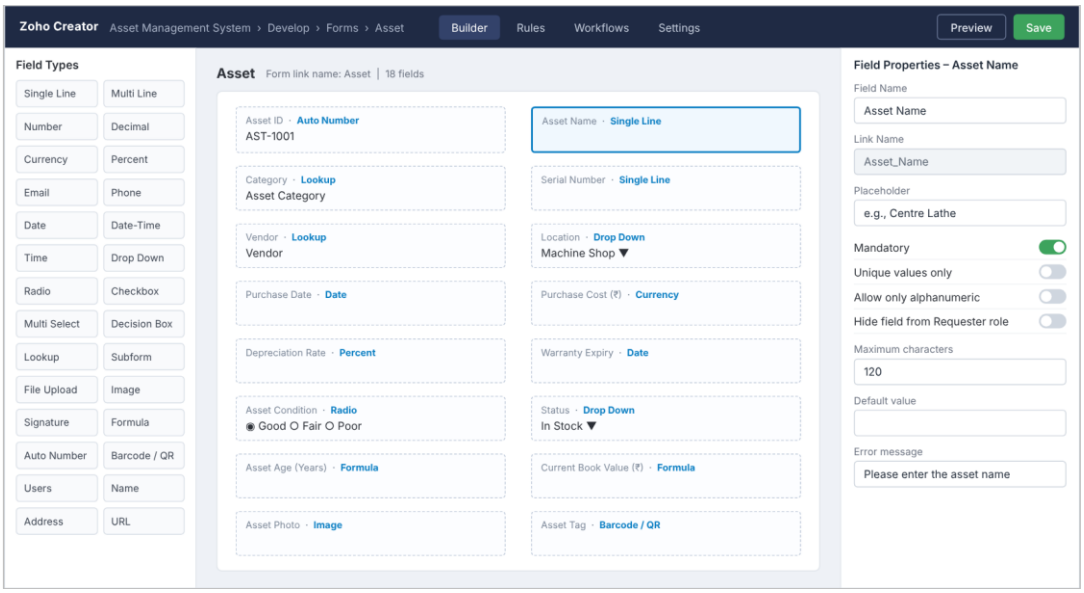


Figure 5.2: Asset Form in the Form Builder

Asset. The register of all assets. The category lookup supplies the depreciation rate; the serial number is unique; the age, book value and warranty status are formula fields shown as read-only; and a photograph and a QR tag can be attached.

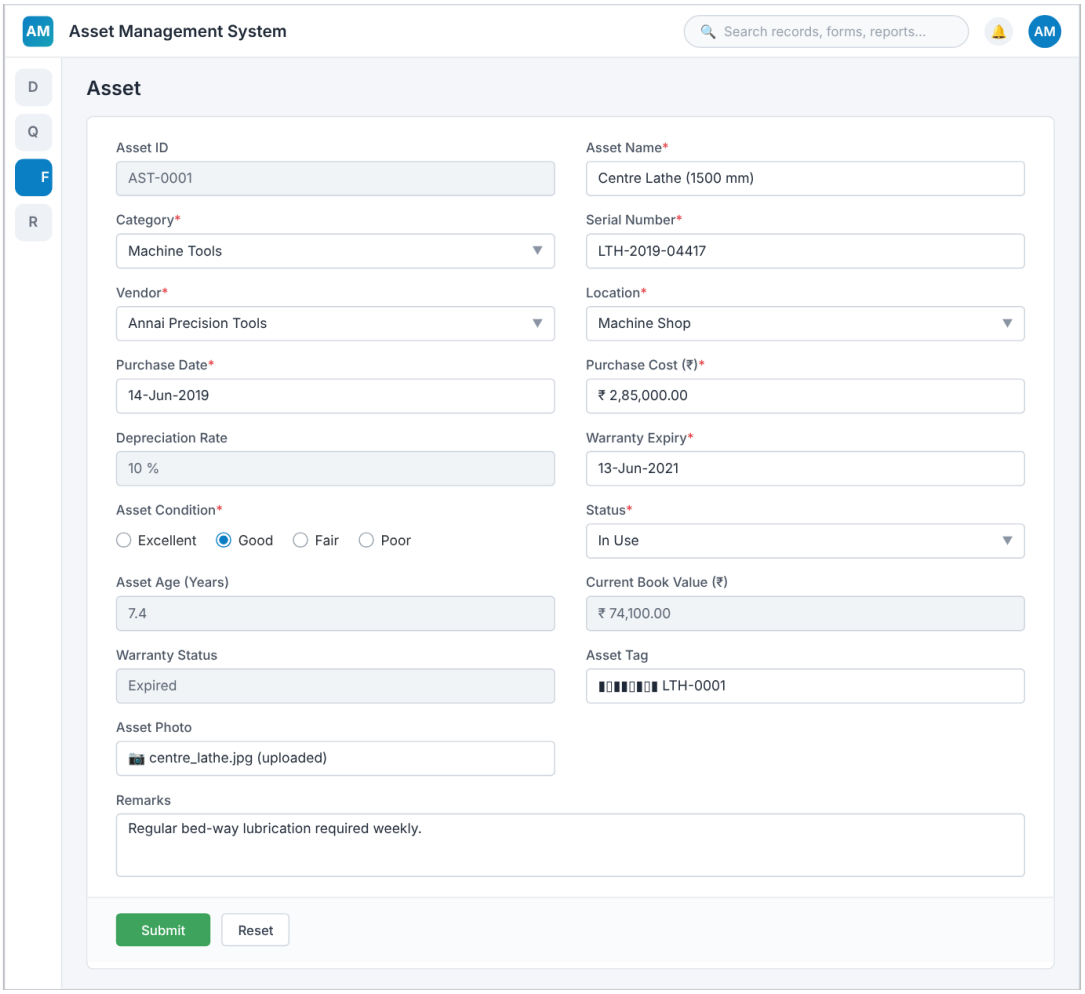


Figure 5.3: Asset Form as Seen by the User

Asset Category. Holds the six categories (Machine Tools, Testing Equipment, Welding Equipment, Metrology Instruments, Computing Devices and Utility Equipment) with depreciation rate and useful life.

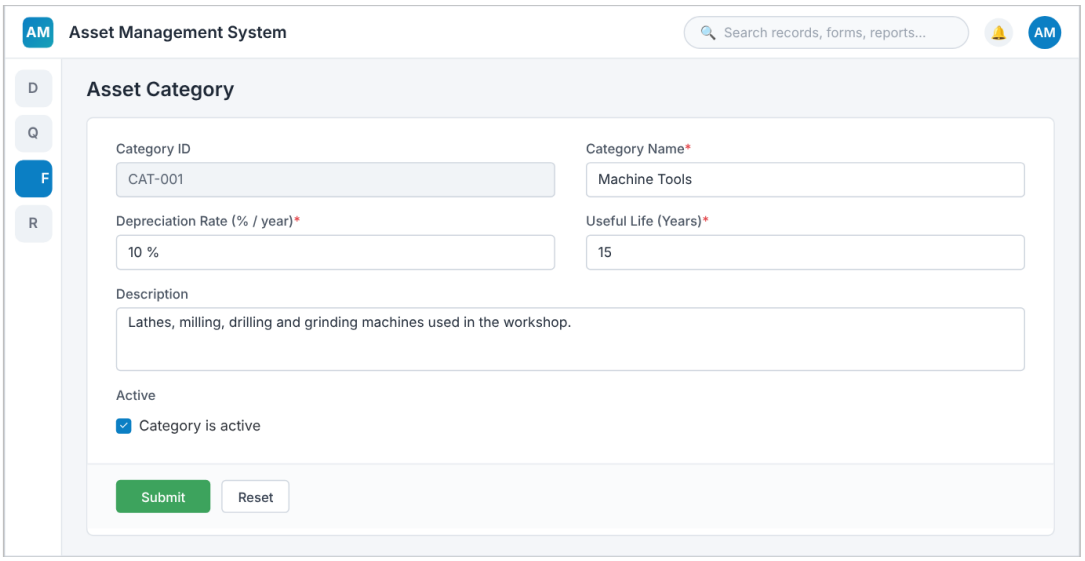


Figure 5.4: Asset Category Form as Seen by the User

Vendor. Holds supplier details with Name, Email, Phone and Address fields, a 15-character GST number validation and a rating.

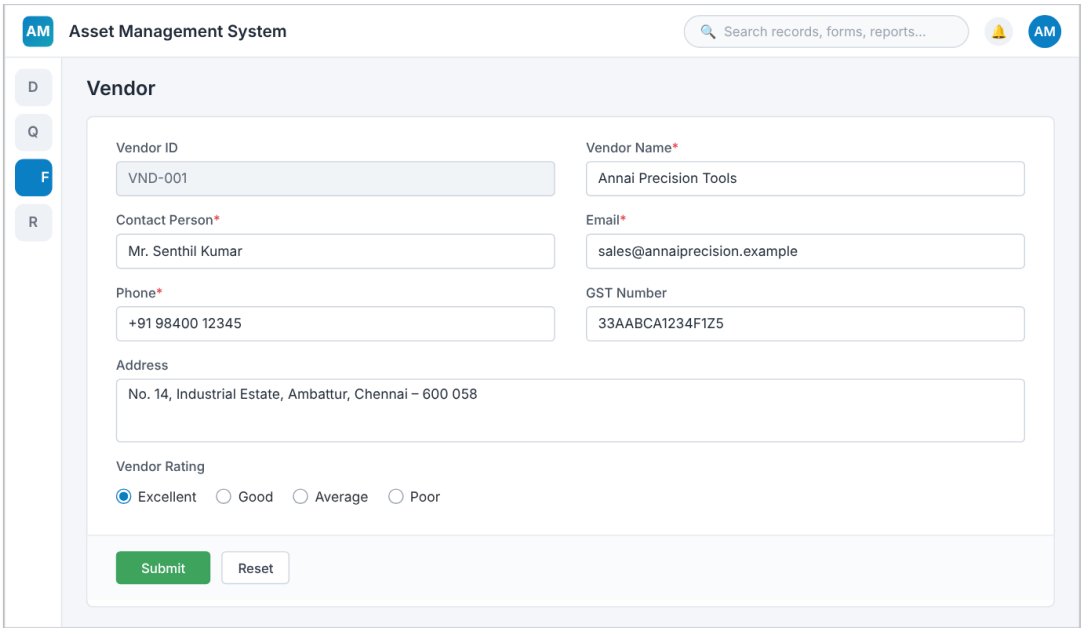


Figure 5.5: Vendor Form as Seen by the User

Asset Request. Used by faculty to ask for new items. The estimated total cost is a formula field, the required-by date cannot be in the past, and the approval status is controlled only by workflow W4.

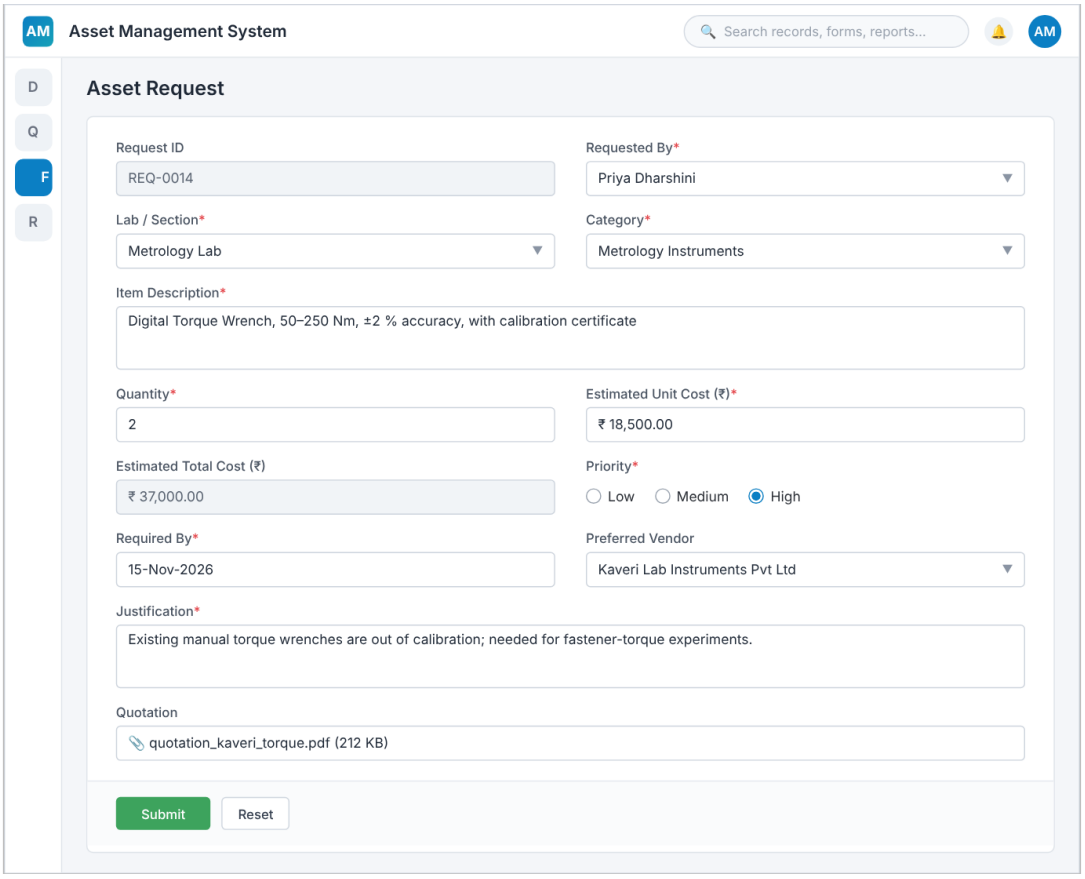


Figure 5.6: Asset Request Form as Seen by the User

Asset Allocation. Records the issue and return of an asset. The asset lookup shows only assets that are In Stock, and the receiver signs on a Signature field.

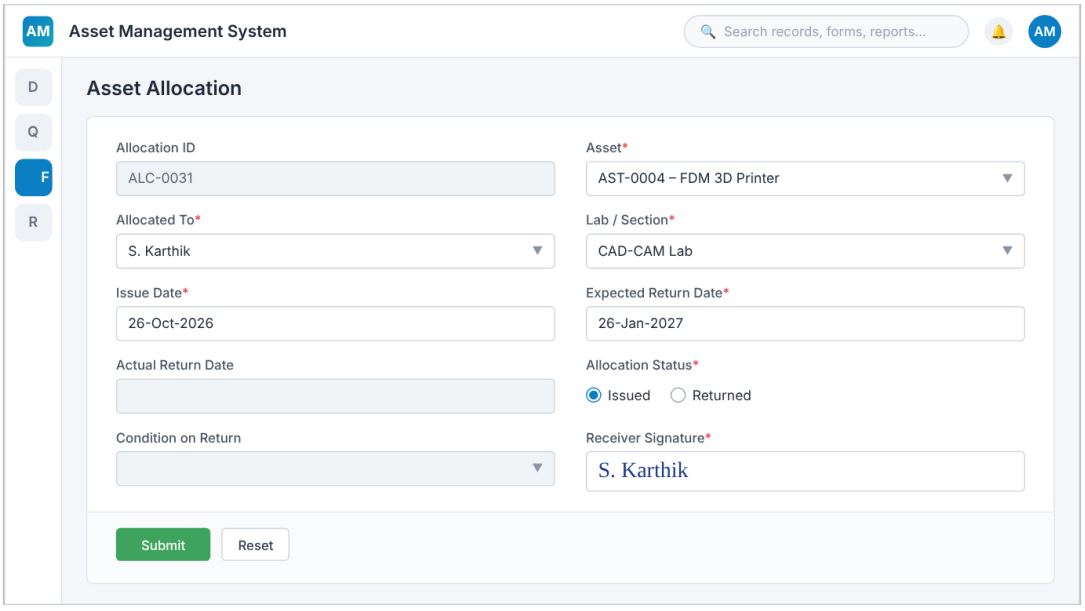


Figure 5.7: Asset Allocation Form as Seen by the User

Maintenance Schedule. Stores the type and frequency of maintenance. The interval in days is set by a field rule and the next due date is a formula field.

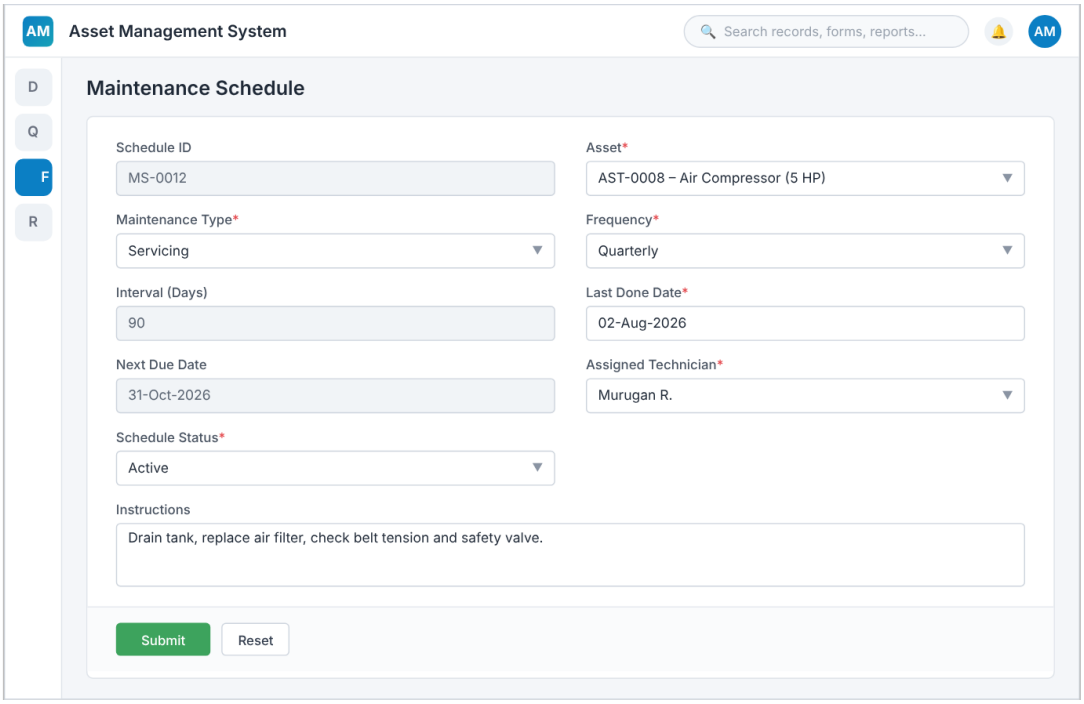


Figure 5.8: Maintenance Schedule Form as Seen by the User

Maintenance Record. The job card. Spare parts are entered in a subform, whose line totals feed the parts cost and total cost formulas. The fault description is shown only for breakdown jobs.

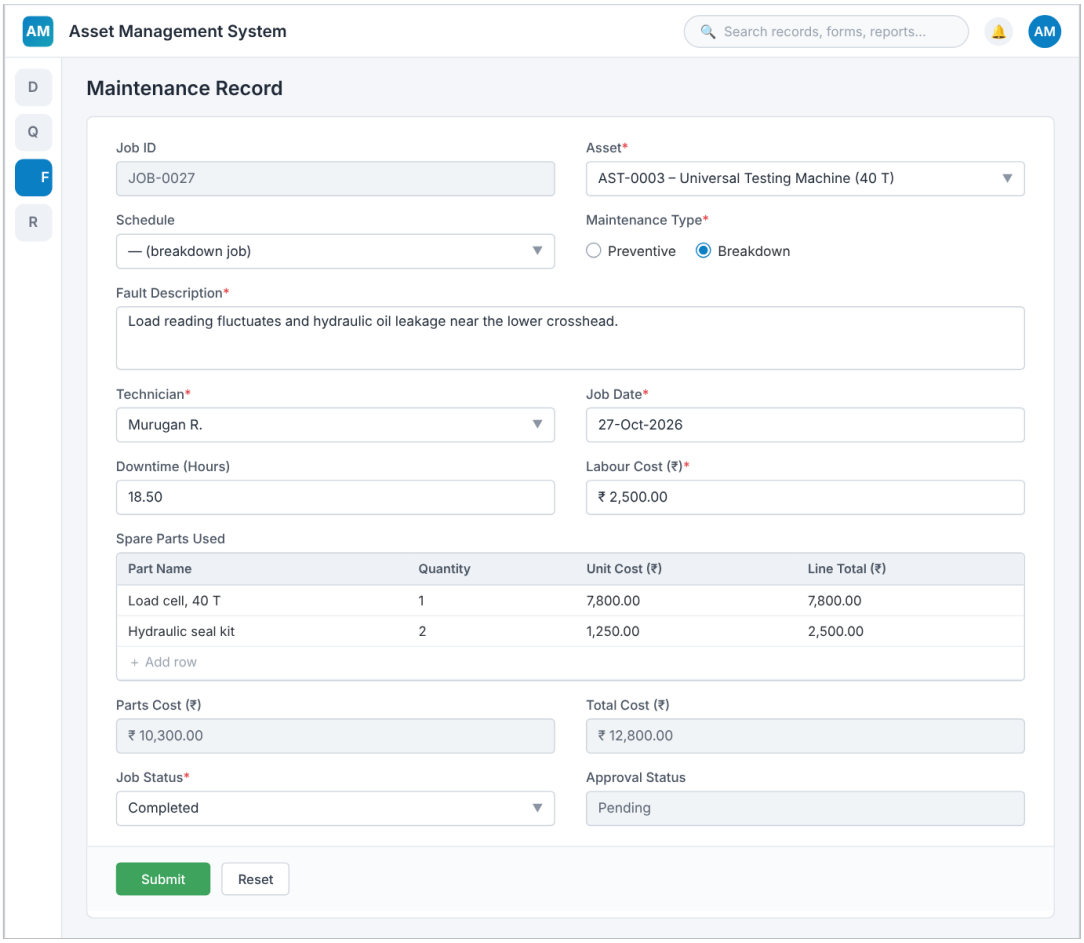


Figure 5.9: Maintenance Record Form as Seen by the User

Asset Disposal. Used to request retirement of an asset. The book value is fetched from the asset, and the request is approved at two levels by workflow W6.

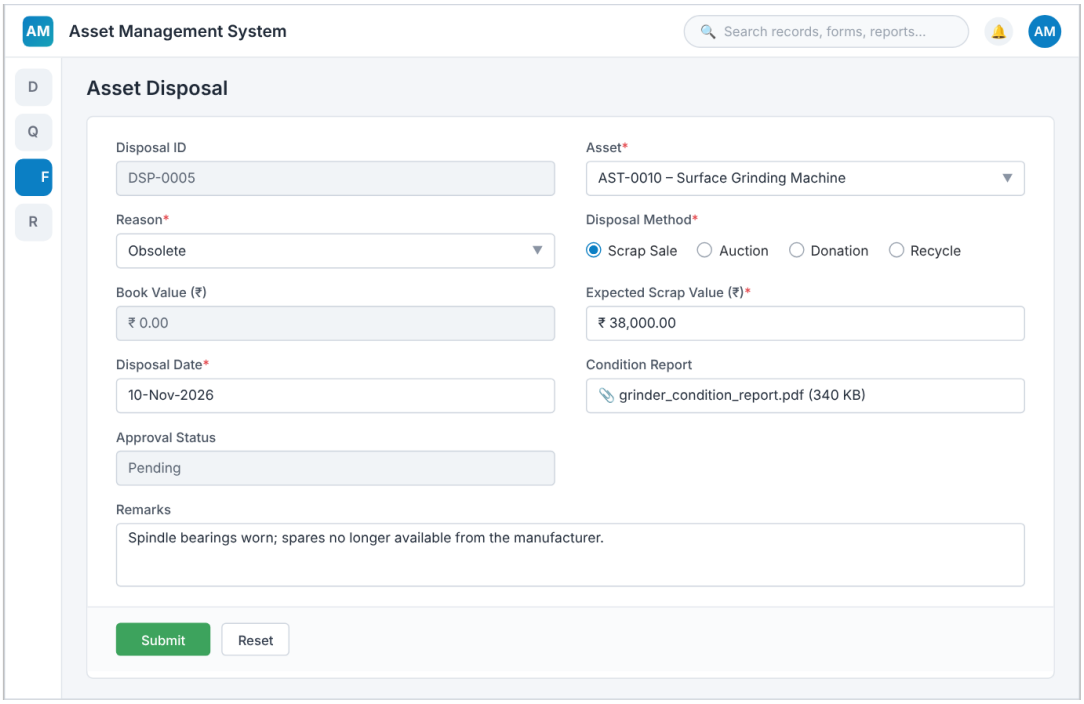


Figure 5.10: Asset Disposal Form as Seen by the User

5.3 Formula Fields and Field Rules

The no-code logic of the application consists of 8 formula fields, 6 field rules and 4 validation rules, listed in Table 5.1. The builder features used follow the feature areas of the Zoho Creator developer certification guide [12]. Formula fields are built with the expression builder by selecting fields and operators; no Deluge script is written. Field rules show, hide, enable or set fields when a value changes, and validation rules stop wrong data from being saved and show an error message.

Table 5.1: Formula Fields and Rules Configured

ID	Type	Form	Condition / Expression	Effect
FM-01	Formula field	Asset	Purchase Date to today ÷ 365	Calculates asset age in years
FM-02	Formula field	Asset	Purchase Cost − (Purchase Cost × Depreciation Rate ÷ 100 × Asset Age)	Calculates current book value
FM-03	Formula field	Asset	If Warranty Expiry < today then “Expired” else “In Warranty”	Shows warranty status
FM-04	Formula field	Asset Request	Quantity × Estimated Unit Cost	Calculates estimated total cost
FM-05	Formula field	Maintenance Schedule	Last Done Date + Interval (Days)	Calculates next due date
FM-06	Formula field	Maintenance Record (subform)	Quantity × Unit Cost	Calculates line total of each part

ID	Type	Form	Condition / Expression	Effect
FM-07	Formula field	Maintenance Record	Sum of Line Total in Spare Parts Used	Calculates parts cost
FM-08	Formula field	Maintenance Record	Labour Cost + Parts Cost	Calculates total cost
RL-01	Field rule	Asset	When Category is selected	Sets Depreciation Rate from the category
RL-02	Field rule	Maintenance Schedule	When Frequency is selected	Sets Interval (Days): 7, 30, 90, 180 or 365
RL-03	Field rule	Asset Request	If Approval Status = Rejected	Shows the Rejection Reason field
RL-04	Field rule	Asset Allocation	If Allocation Status = Returned	Enables Actual Return Date and shows Condition on Return
RL-05	Field rule	Maintenance Record	If Maintenance Type = Breakdown	Shows Fault Description (mandatory)
RL-06	Field rule	Asset Disposal	When Asset is selected	Sets Book Value from the asset
VL-01	Validation rule	Asset	Purchase Date ≤ today	Error: purchase date cannot be a future date
VL-02	Validation rule	Vendor	GST Number has 15 characters	Error: enter a valid 15-character GST number
VL-03	Validation rule	Asset Request	Required By ≥ today	Error: required-by date cannot be in the past
VL-04	Validation rule	Asset Allocation	Expected Return Date ≥ Issue Date	Error: return date is before issue date

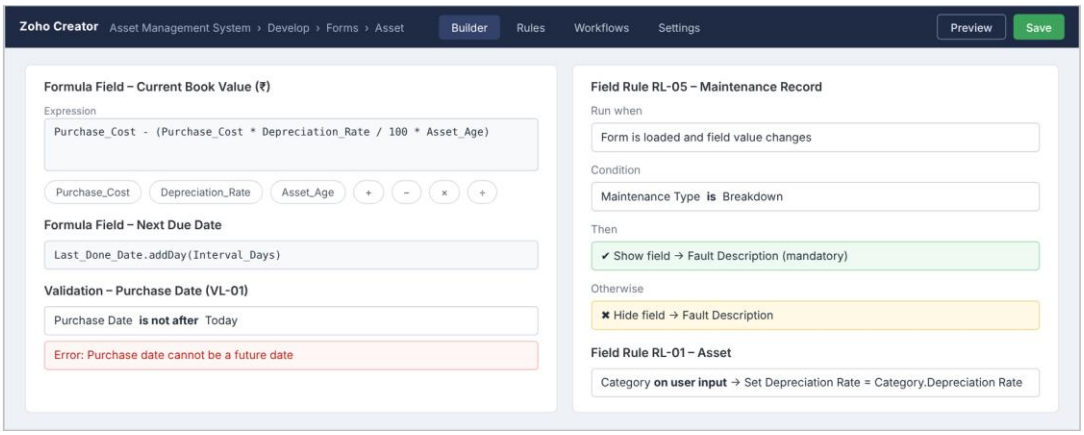


Figure 5.11: Field Rule and Formula Configuration

5.4 Reports

Nine reports of six different types were configured, as listed in Table 5.2. Filters, grouping, sorting, conditional formatting and record-level restrictions were set in the report builder. Figures 5.12 to 5.17 show the main reports with the sample data.

Table 5.2: Reports Configured

Report Name	Type	Based on Form	Filter / Group / Sort	Formatting / Buttons	Roles
All Assets	List	Asset	Group by Category; sort by Asset ID	Red text if Warranty Expired; amber if Under Maintenance	Admin, Supv., Tech.
Allocation Register	List	Asset Allocation	Filter: Allocation Status = Issued	Red row if Expected Return Date has passed	Admin, Supv.
Purchase Requests	List	Asset Request	Group by Approval Status	Green / amber / red status badge; Approve button	Admin, Supv., Req. (own)
Disposal Register	List	Asset Disposal	Sort by Disposal Date	Status badge; export to PDF button	Admin, Supv.
Maintenance Job Board	Kanban	Maintenance Record	Columns by Job Status	Card shows Asset, Technician, Total Cost	Admin, Supv., Tech. (assigned)
Maintenance Calendar	Calendar	Maintenance Schedule	Date field: Next Due Date	Colour by Maintenance Type	Admin, Supv., Tech. (assigned)
Asset Value by Category	Chart (Bar)	Asset	Group by Category; sum of Book Value	Data labels shown	Admin, Supv.
Maintenance Cost Pivot	Pivot	Maintenance Record	Rows: Category; Columns: Month; Value: sum of Total Cost	Grand totals shown	Admin, Supv.
Vendor Directory	Spreadsheet	Vendor	Sort by Vendor Name	Inline editing for Administrator	Admin

The **All Assets** report groups assets by category and shows the warranty status in red when expired and the status as a coloured badge.

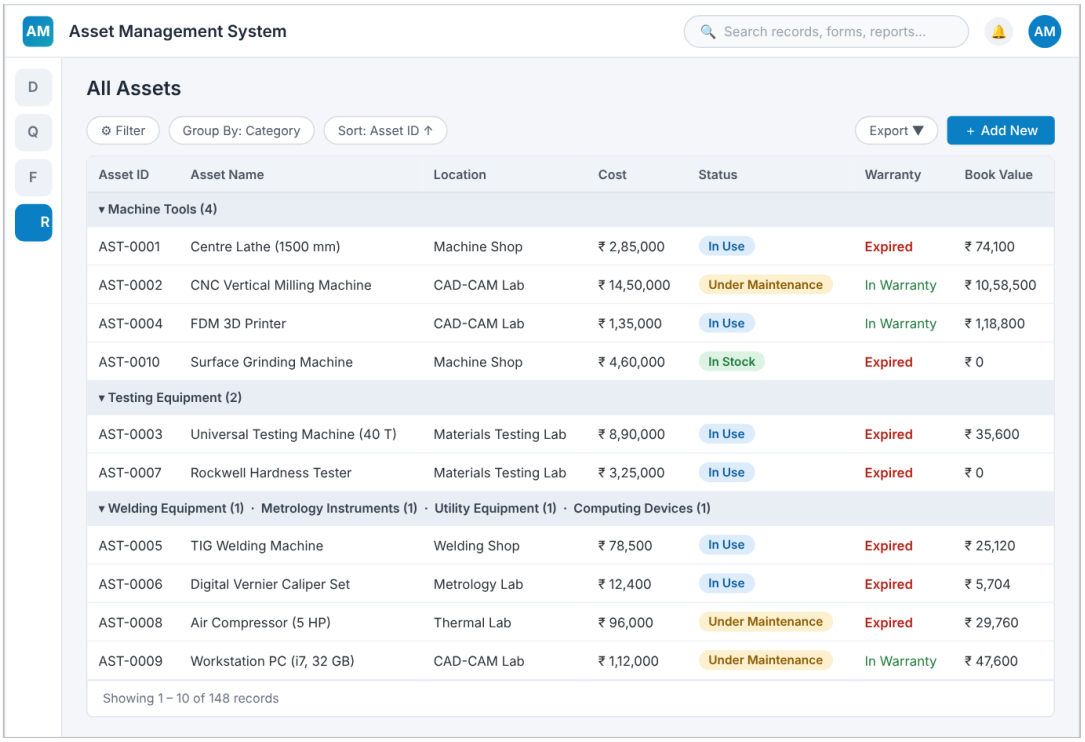


Figure 5.12: All Assets Report

The **Purchase Requests** report shows each request with a coloured approval-status badge; approvers see Approve and Reject buttons for pending items.

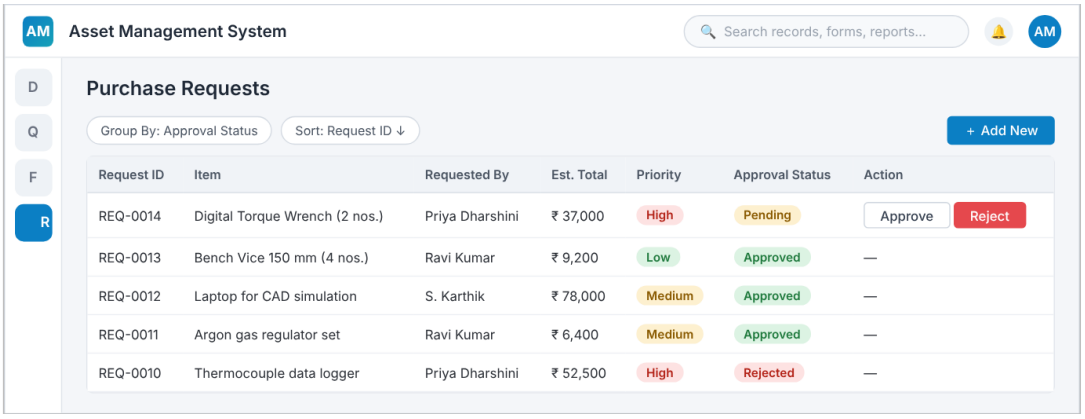


Figure 5.13: Purchase Requests Report

The **Maintenance Job Board** is a Kanban report with columns for Open, In Progress and Completed jobs; each card shows the asset, technician and total cost.

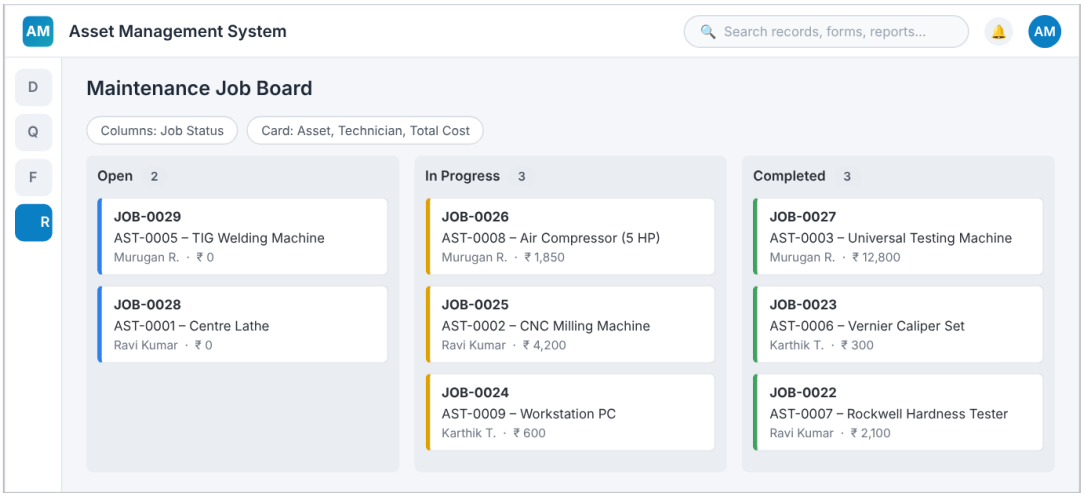


Figure 5.14: Maintenance Job Board (Kanban) Report

The **Maintenance Calendar** plots the next due date of every schedule and colours each entry by maintenance type.

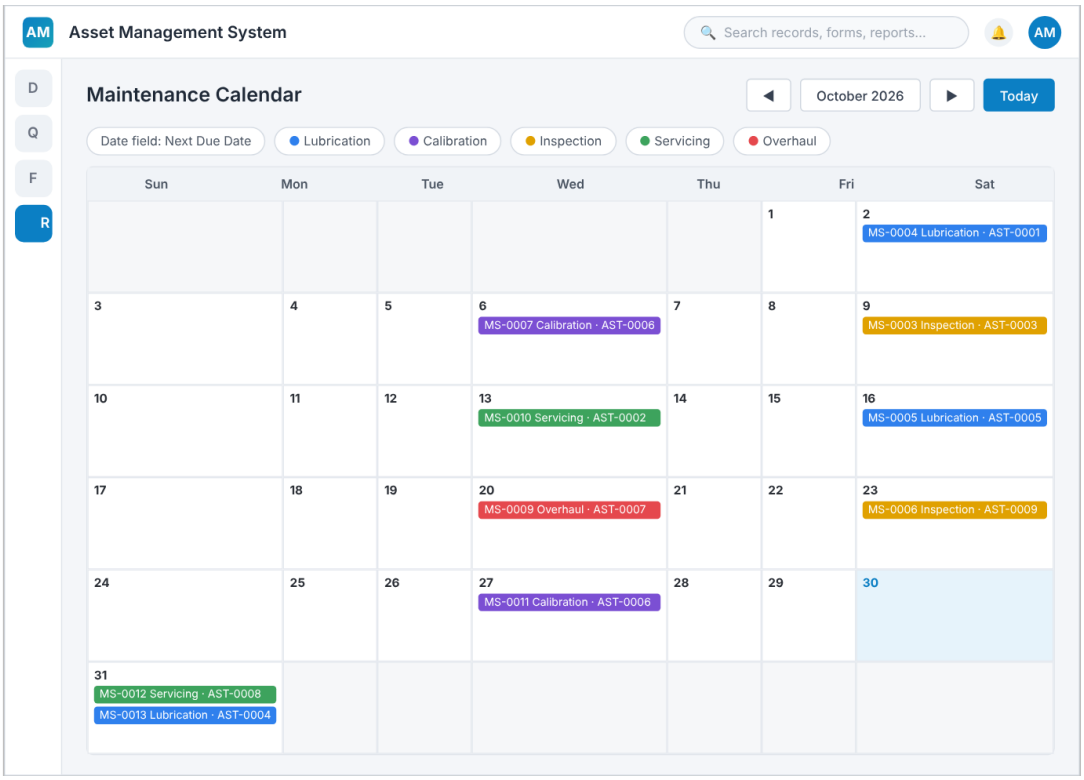


Figure 5.15: Maintenance Calendar Report

The **Asset Value by Category** bar chart sums the current book value for each category.

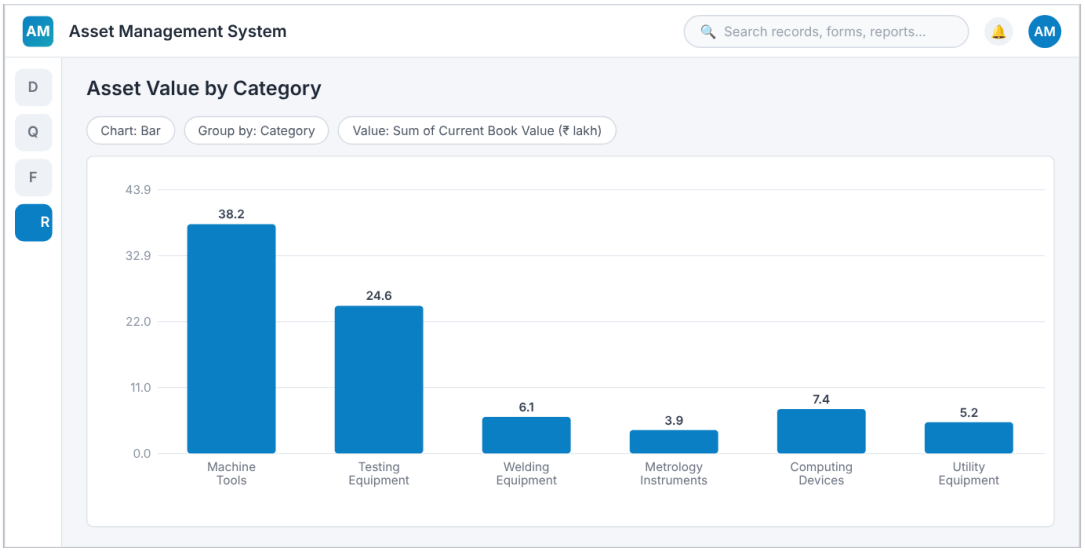
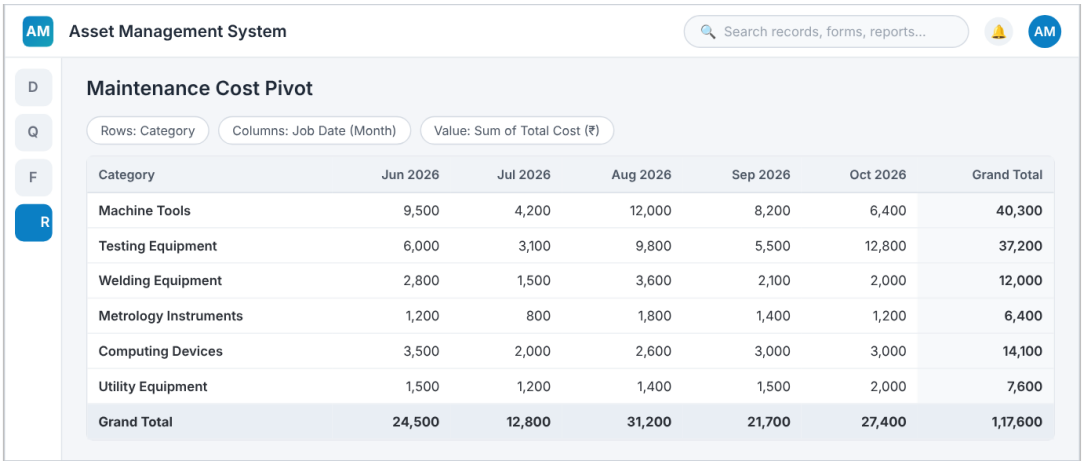


Figure 5.16: Asset Value by Category (Chart) Report

The **Maintenance Cost Pivot** shows the total repair and servicing cost by category and month, with grand totals.



Category	Jun 2026	Jul 2026	Aug 2026	Sep 2026	Oct 2026	Grand Total
Machine Tools	9,500	4,200	12,000	8,200	6,400	40,300
Testing Equipment	6,000	3,100	9,800	5,500	12,800	37,200
Welding Equipment	2,800	1,500	3,600	2,100	2,000	12,000
Metrology Instruments	1,200	800	1,800	1,400	1,200	6,400
Computing Devices	3,500	2,000	2,600	3,000	3,000	14,100
Utility Equipment	1,500	1,200	1,400	1,500	2,000	7,600
Grand Total	**24,500**	**12,800**	**31,200**	**21,700**	**27,400**	**1,17,600**

Figure 5.17: Maintenance Cost (Pivot) Report

5.5 Pages and Dashboards

Two pages were built with the page builder. The **Asset Dashboard** is shown to the Administrator and the Supervisor. It has six KPI tiles (total assets, book value, assets under maintenance, pending approvals, warranties expiring in 30 days and overdue maintenance), a bar chart of assets by category, a donut chart of assets by status, a line chart of monthly maintenance cost and a table of upcoming maintenance. The **Quick Actions** page is shown to all roles and contains buttons that open the most-used forms, an embedded report of the user's own pending requests and an embedded quick-request form.

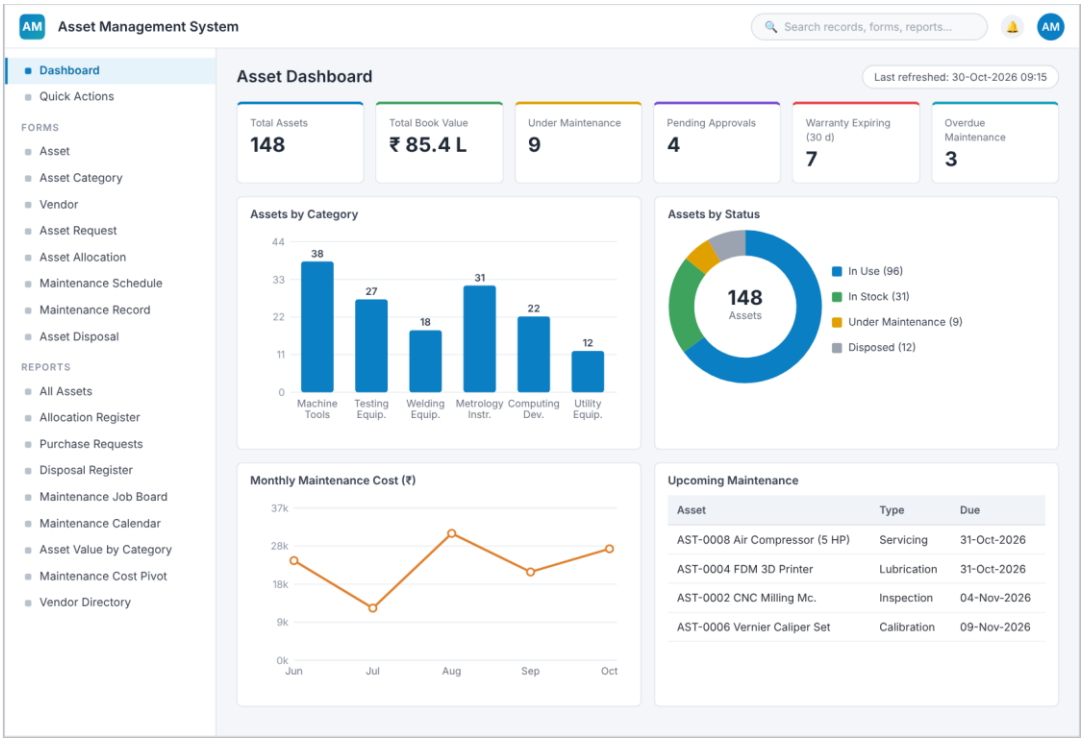


Figure 5.18: Asset Dashboard Page

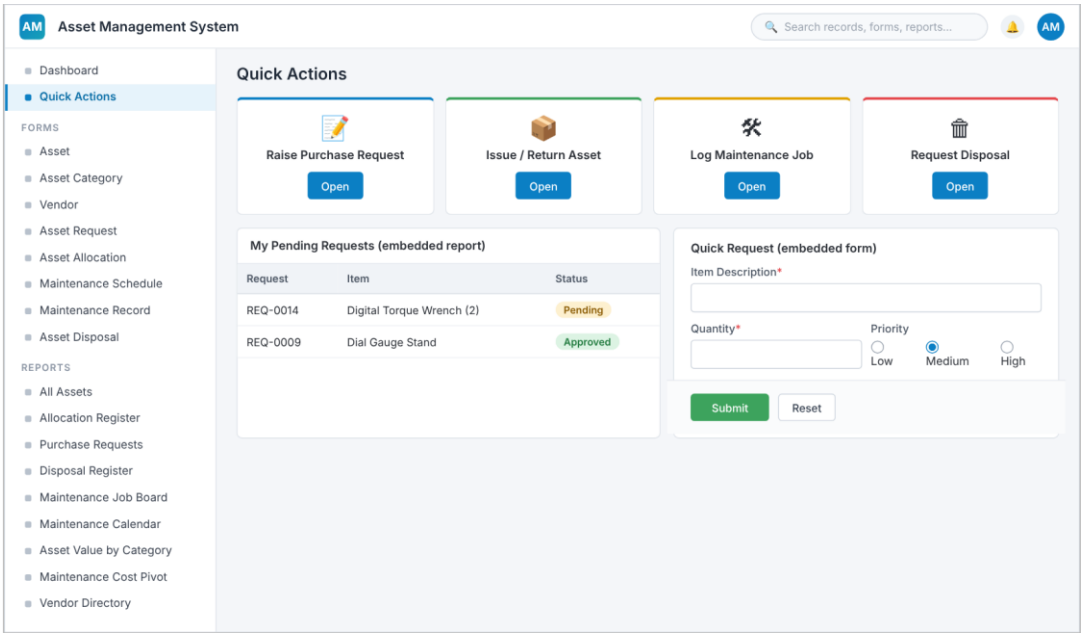


Figure 5.19: Quick Actions Page with Embedded Form and Report

5.6 Workflows and Automation

Eight workflows were configured with the visual workflow builder without any scripting: three form workflows (W1–W3), three approval processes (W4–W6) and two schedules (W7–W8). They are summarised in Table 5.3. Blueprint was not used because the stage-by-stage life of an asset is already controlled by the Status field and the approval processes.

Table 5.3: Workflows Configured

ID	Workflow Name	Type	Trigger	Condition	Action(s)
W1	Asset Registration Alert	Form workflow	On Add → On Success	—	Send e-mail to Administrator; send push notification to Supervisor
W2	Allocation Status Updater	Form workflow	On Add / On Edit → On Success	Allocation Status = Issued or Returned	Update Asset Status (In Use / In Stock); set Allocation Status = Returned when Actual Return Date is filled
W3	Maintenance Completion Updater	Form workflow	On Add / On Edit → On Success	Job Status = Completed	Update Last Done Date in Maintenance Schedule; restore Asset Status; e-mail Supervisor with Total Cost
W4	Purchase Request Approval	Approval	On Add	Estimated Total Cost > ₹25,000 needs Level 2	Level 1: Administrator; Level 2: Supervisor; update Approval Status; e-mail Requester
W5	Repair Cost Approval	Approval	On Add / On Edit	Total Cost > ₹10,000	Supervisor approves or rejects; update Approval Status; e-mail Technician
W6	Asset Disposal Approval	Approval	On Add	All disposal requests	Level 1: Administrator; Level 2: Supervisor; on approval set Asset Status = Disposed
W7	Preventive Maintenance Reminder	Schedule	Daily at 08:00 AM	Next Due Date ≤ today + 3 days and Schedule Status = Active	E-mail and push notification to Assigned Technician; e-mail Supervisor if overdue
W8	Warranty Expiry Alert	Schedule	Weekly, Monday 09:00 AM	Warranty Expiry within 30 days and Status ≠ Disposed	E-mail list of expiring assets to Administrator

W1 – Asset Registration Alert (form workflow). When an asset is added, an e-mail is sent to the Administrator and a push notification to the Supervisor, so that new assets are noticed immediately.

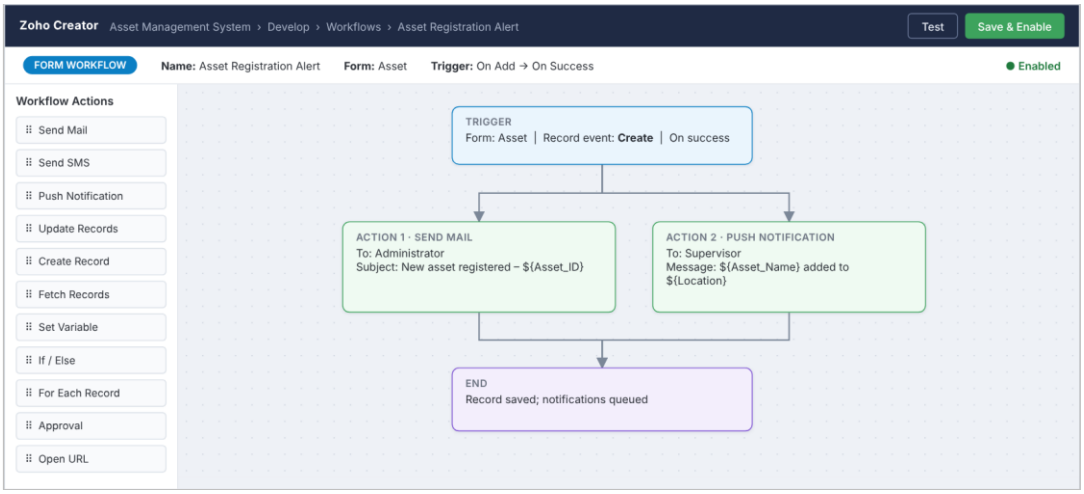


Figure 5.20: W1 – Asset Registration Alert in the Workflow Builder

W2 – Allocation Status Updater (form workflow). When an allocation is added or edited, the asset status is set to In Use for an issue, or the allocation is marked Returned and the asset set to In Stock when a return date is entered.

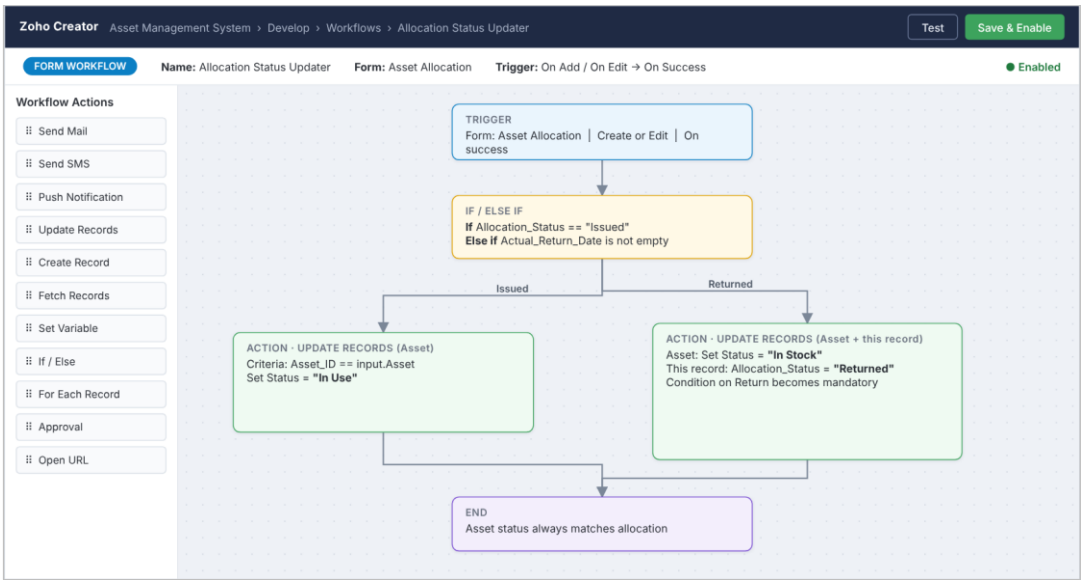


Figure 5.21: W2 – Allocation Status Updater in the Workflow Builder

W3 – Maintenance Completion Updater (form workflow). When a job is completed, the last-done date of the linked schedule is updated, the next due date recalculates, the asset status is restored and the Supervisor is e-mailed with the total cost. For any other job status the asset is marked Under Maintenance.

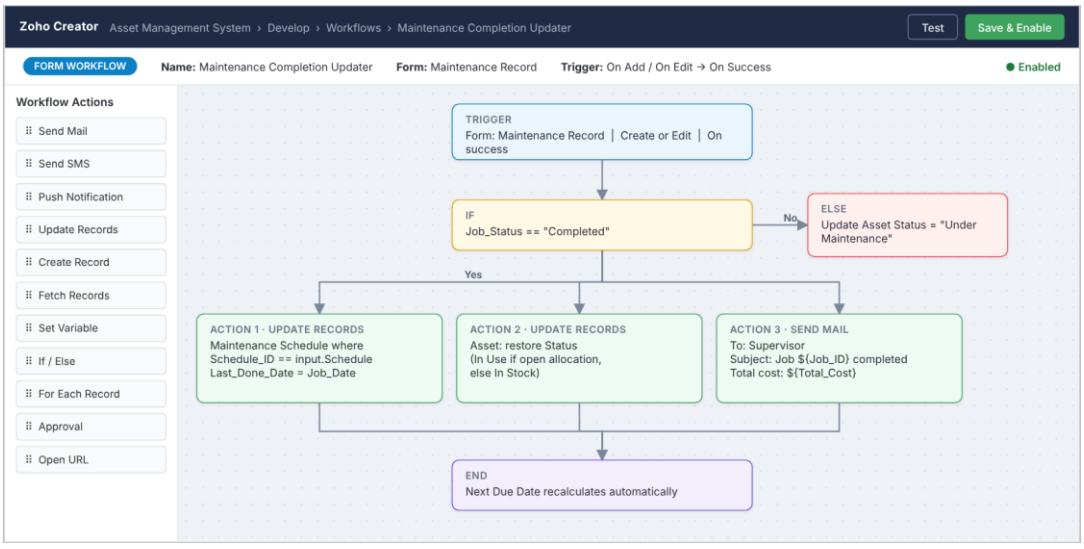


Figure 5.22: W3 – Maintenance Completion Updater in the Workflow Builder

W4 – Purchase Request Approval (approval). Every request goes to the Administrator at Level 1; if the estimated total cost exceeds ₹25,000 it goes on to the

Supervisor at Level 2. The approval status is updated and the requester is informed (Figure 5.24 shows the approval settings).

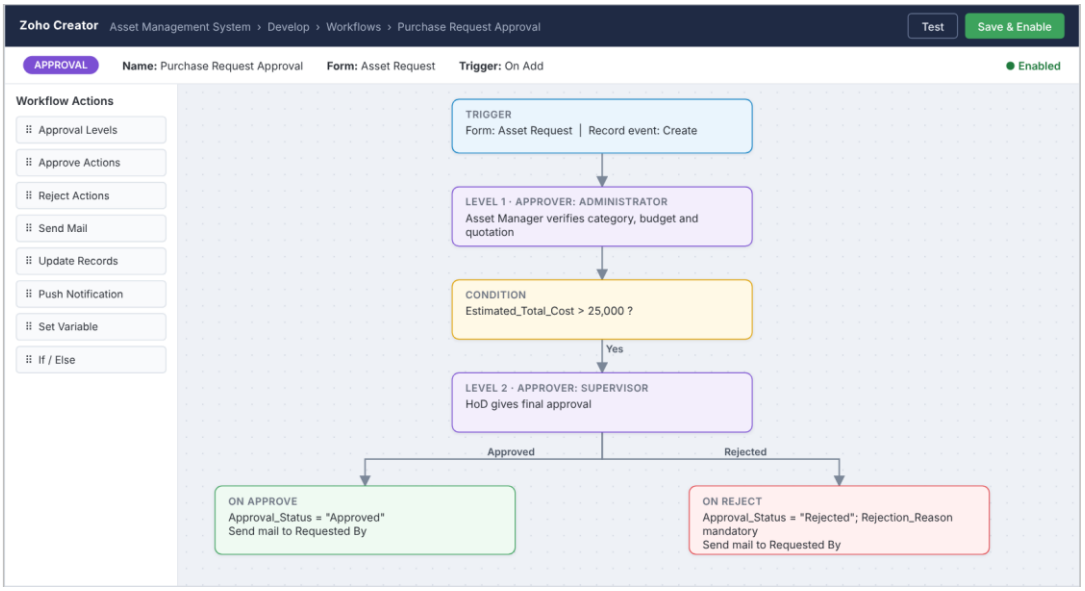


Figure 5.23: W4 – Purchase Request Approval in the Workflow Builder

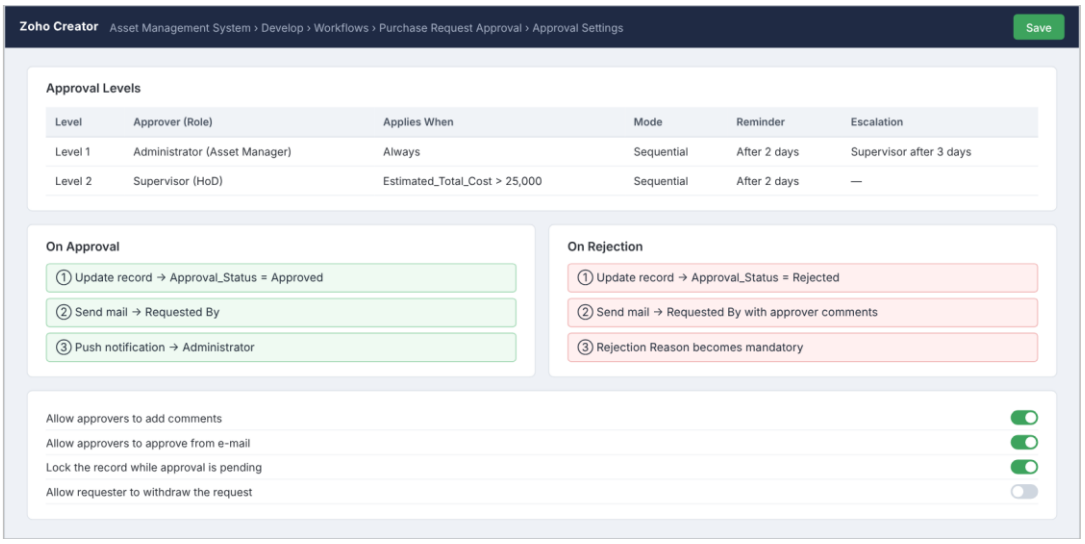


Figure 5.24: Approval Configuration for Purchase Request Approval

W5 – Repair Cost Approval (approval). A completed job whose total cost is above ₹10,000 is sent to the Supervisor. If it is rejected, the job returns to In Progress and the technician is e-mailed with the comments.

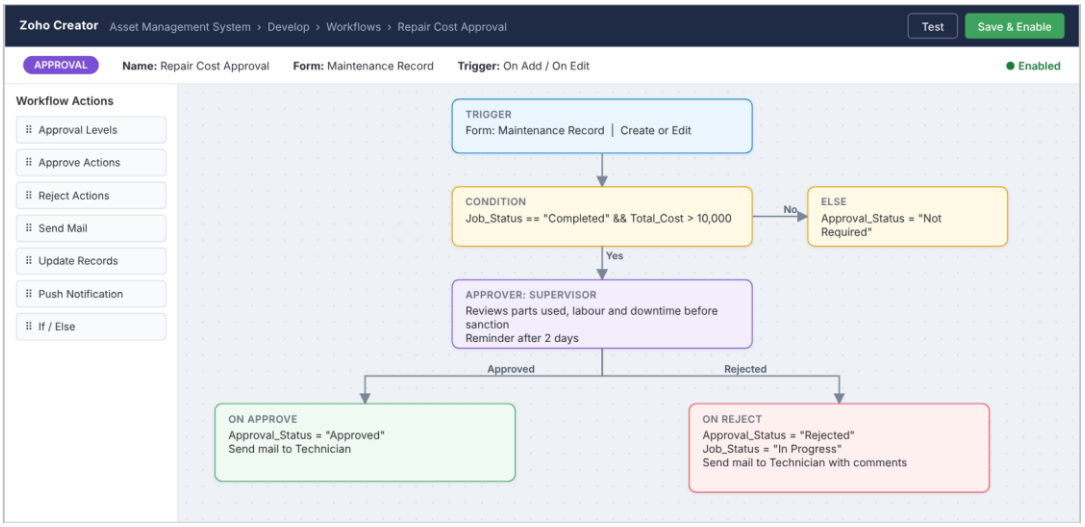


Figure 5.25: W5 – Repair Cost Approval in the Workflow Builder

W6 – Asset Disposal Approval (approval). A disposal request needs approval from both the Administrator and the Supervisor; on approval the asset status becomes Disposed.

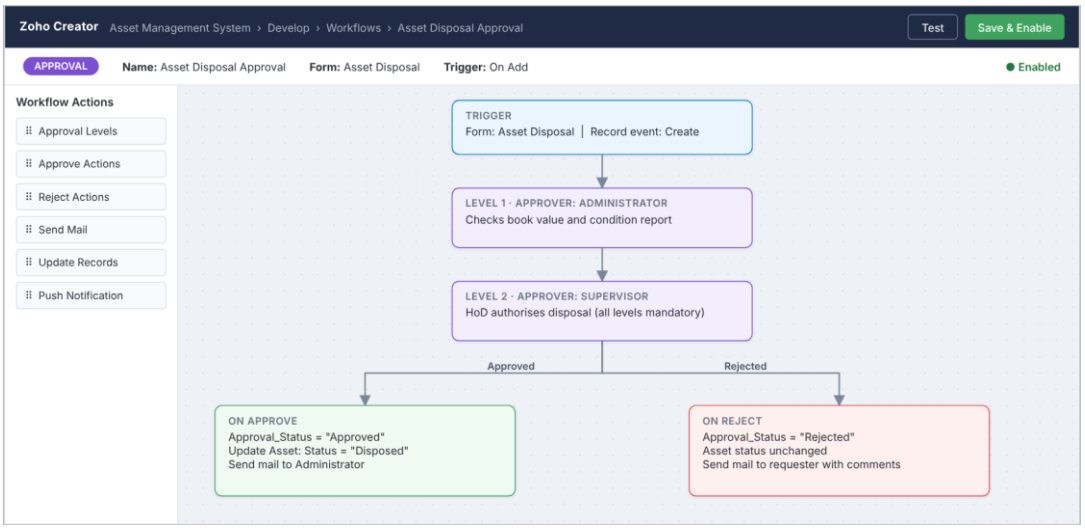


Figure 5.26: W6 – Asset Disposal Approval in the Workflow Builder

W7 – Preventive Maintenance Reminder (schedule). Every day at 08:00 AM the schedule finds active maintenance plans due within three days and sends an e-mail and a push notification to the assigned technician; overdue plans are also e-mailed to the Supervisor.

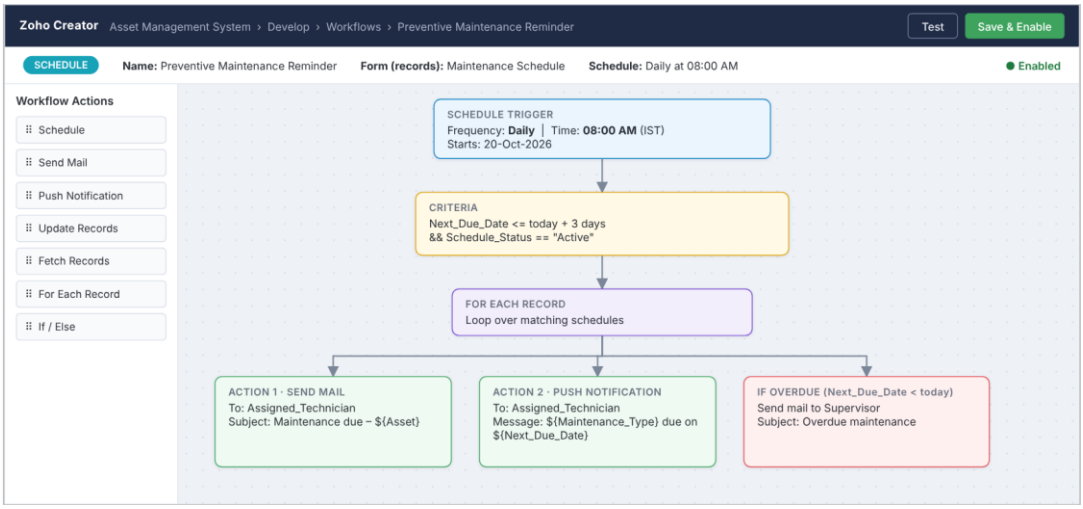


Figure 5.27: W7 – Preventive Maintenance Reminder in the Workflow Builder

W8 – Warranty Expiry Alert (schedule). Every Monday at 09:00 AM it collects assets whose warranty expires within 30 days and e-mails the list to the Administrator.

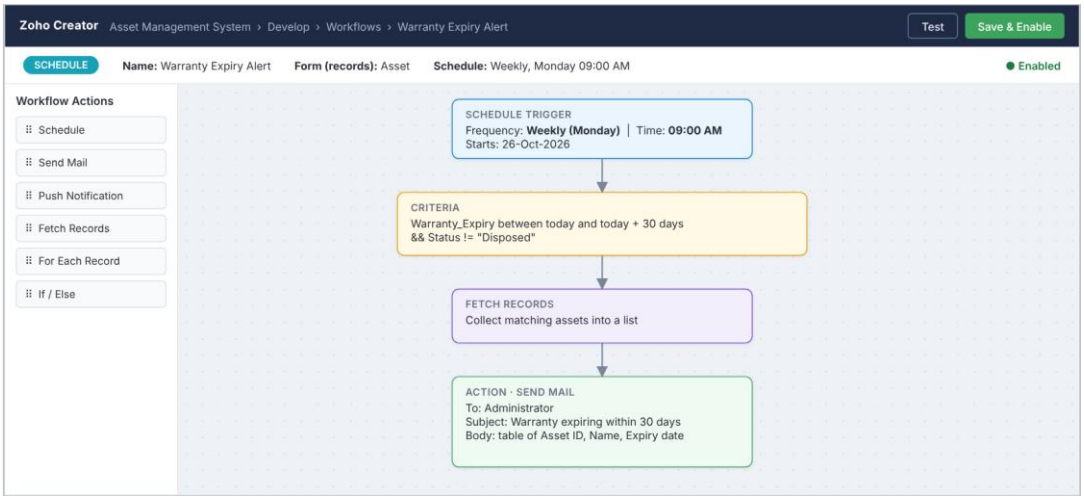


Figure 5.28: W8 – Warranty Expiry Alert in the Workflow Builder

5.7 Users, Roles and Permissions

The permission matrix of Section 4.6 was implemented under Settings → Users & Control. Six demonstration users were invited and assigned to four roles. For each role a permission set defines which forms, reports and pages can be viewed, added, edited or deleted. Record-level rules limit a technician to the maintenance records and schedules assigned to him or her, and a requester to the requests and disposals that he or she created. Cost fields on the Asset Request form are hidden from users outside the Administrator and Supervisor roles. Figure 5.29 shows the user list and the Technician permission set.

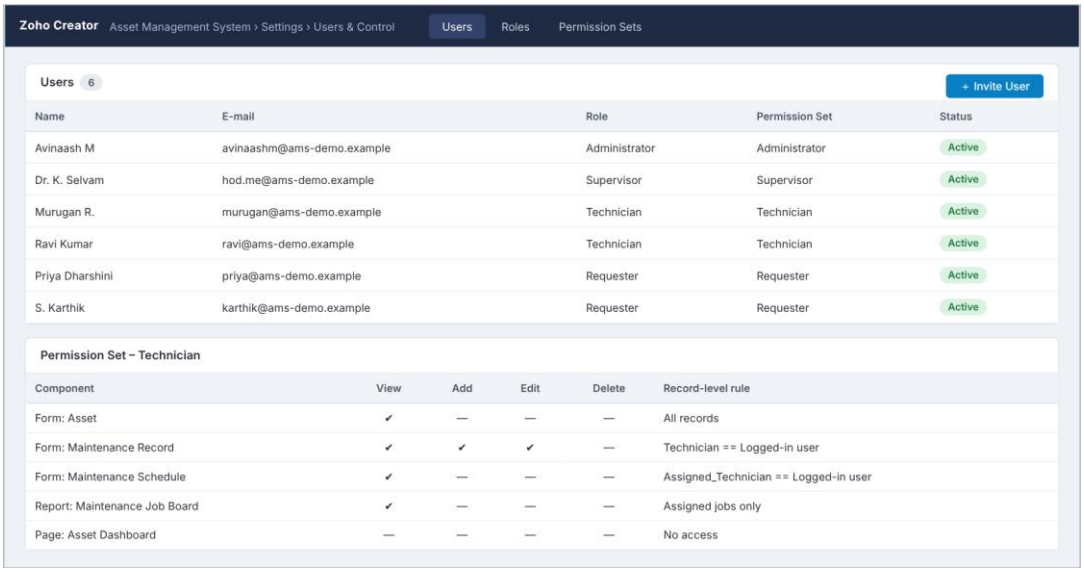


Figure 5.29: Roles and Permissions Configuration

5.8 Data Import, Export and Integration

The master data of 150 assets was kept in an Excel sheet. It was imported into the Asset form using the Import Data option: the sheet columns were mapped to form fields, the lookup values were matched to the categories and vendors, and records with a duplicate serial number were skipped (Figure 5.30). Reports can be exported to PDF, Excel and CSV from the Export menu, and a read-only link can be generated for a report if public access is needed (it is disabled in this application). Notification e-mails are sent through Zoho Mail, and report exports open in Zoho Sheet; both are built-in integrations that required only configuration.

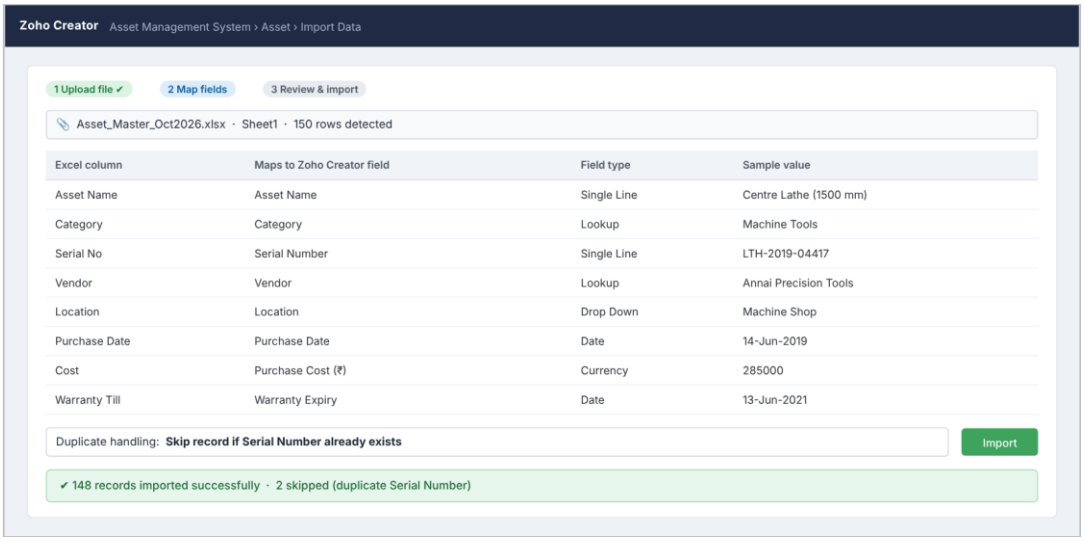


Figure 5.30: Data Import Setup for the Asset Form

5.9 AI-Assisted Features

The Zia AI application builder was used once, at the start of the project. The prompt "An asset management system for a mechanical engineering workshop to register machines, track purchase requests with approval, allocate assets to labs, schedule preventive maintenance and record repair costs" produced four draft forms (Asset, Maintenance, Employee and Vendor). The author kept the Asset and Vendor forms and added about ten fields to the Asset form, split Maintenance into Schedule and Record, and removed the Employee form because the Users field type serves the same purpose. All other forms, rules, reports, pages and workflows were created manually. Figure 5.31 shows the AI builder with the suggestions and the author's actions.

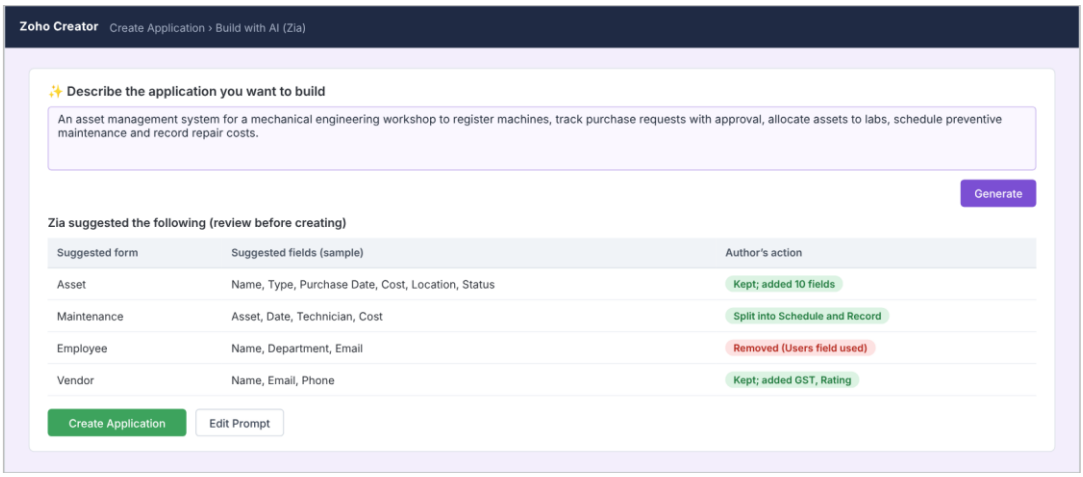


Figure 5.31: Zia AI Application Builder Used for the First Draft

5.10 Mobile Application

The application was opened on the Zoho Creator mobile app after logging in with the same credentials. The menu, forms and reports appear automatically in a mobile layout. In the Asset form, the Asset Tag field opens the camera to scan a barcode or QR code, and the Asset Photo field captures a picture. Push notifications for maintenance reminders, pending approvals and warranty alerts arrive on the phone, and approvers can approve or reject a request directly from the notification (Figure 5.32).

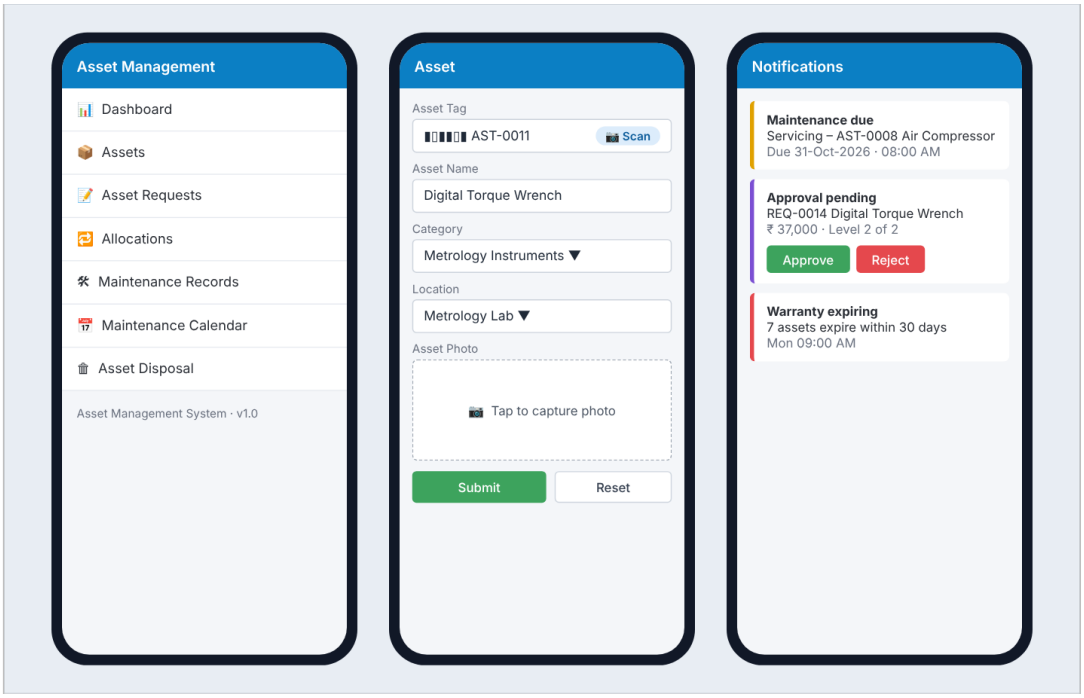


Figure 5.32: Application on the Zoho Creator Mobile App

CHAPTER 6
TESTING

6.1 Testing Approach

Testing was carried out in the Development environment before publishing. Each form was tested with correct inputs, with a mandatory field left empty and with wrong values (for example a future purchase date or a 12-character GST number). Each workflow was triggered by adding or editing a record, and its e-mails, push notifications and field updates were checked. The two schedules were tested by setting the run time a few minutes ahead and confirming that only matching records were selected. Each report and the dashboard were checked against the stored data, and the application was opened with a user of each role to confirm that permissions and record-level rules work. Test records were deleted before publishing.

6.2 Test Cases

Twenty test cases were written to cover every functional requirement of Table 3.2 and the main non-functional requirements, including negative tests for wrong input and unauthorised access. The results are in Table 6.1.

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Register an asset with all mandatory fields	Add AST-0001 with all fields	Record saved; success message	Saved; ID AST-0001 generated	Pass
TC-02	FR-01	Submit Asset form with Serial Number empty	Leave Serial Number blank	Error shown; not saved	Error “Enter Serial Number”	Pass
TC-03	FR-01	Add an asset with a duplicate serial number	Repeat serial LTH-2019-04417	Duplicate rejected	Error “Value already exists”	Pass
TC-04	FR-01	Enter a future purchase date	Purchase Date = 31-Dec-2027	Validation VL-01 error	Error shown; not saved	Pass
TC-05	FR-02	Check calculated age and book value	Cost ₹2,85,000; rate 10 %; age 7.4	Book value ₹74,100	₹74,100.00 shown	Pass
TC-06	FR-03	Add a category with an existing name	Machine Tools again	Duplicate rejected	Error shown; not saved	Pass
TC-07	FR-04	Raise a request and check total	Qty 2 × ₹18,500	Total ₹37,000	₹37,000.00 shown	Pass
TC-08	FR-05	Request above ₹25,000 is approved at two levels	REQ-0014, ₹37,000	Level 1 then Level 2; status Approved	Both levels shown; status Approved	Pass
TC-09	FR-05	Reject a request	Reject REQ-0010 with comment	Status Rejected; reason shown; e-mail sent	As expected	Pass

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-10	FR-06	Allocate an asset	Issue AST-0004 to CAD-CAM Lab	Asset status In Use	Status changed to In Use	Pass
TC-11	FR-06	Return an allocated asset	Enter actual return date	Allocation Returned; asset In Stock	As expected	Pass
TC-12	FR-07	Next due date calculation	Last done 02-Aug-2026; Quarterly	Next due 31-Oct-2026	31-Oct-2026 shown	Pass
TC-13	FR-08	Reminder before due date	Run W7 for due date in 3 days	Technician e-mailed	E-mail and push received	Pass
TC-14	FR-09	Breakdown job shows fault field	Select Breakdown	Fault Description visible and mandatory	Field displayed	Pass
TC-15	FR-09	Subform and cost totals	Parts 7,800 + 2,500; labour 2,500	Total ₹12,800	₹12,800.00 shown	Pass
TC-16	FR-10, FR-11	Completed job above ₹10,000	JOB-0027 Completed	Sent to Supervisor; schedule updated	Approval Pending; schedule updated	Pass
TC-17	FR-12, FR-13	Disposal approval and warranty alert	DSP-0005; run W8	Asset Disposed; alert e-mailed	As expected	Pass
TC-18	FR-15, NFR-02	Requester opens the Dashboard	Login as Requester	Access denied; menu hidden	Dashboard not visible	Pass
TC-19	FR-14	Dashboard and report totals	Compare KPI with All Assets count	Both show 148 assets	148 on both	Pass
TC-20	FR-16, NFR-03	Use the mobile app	Scan asset tag and add photo	Tag read; photo saved	Tag and photo saved	Pass

6.3 Problems Faced and Fixes

Four problems were found during testing and fixed. They are recorded in Table 6.2.

Table 6.2: Issue Log

Issue ID	Problem Observed	Cause	Fix Applied	Status
ISS-01	Asset Allocation lookup listed assets that were already In Use	Lookup showed all assets	Added a lookup filter: Status = In Stock	Closed
ISS-02	Book value became negative for very old assets	Formula did not stop at zero	Changed FM-02 to return 0 when the result is negative	Closed
ISS-03	Technician could see all maintenance records	Record-level rule was missing	Added rule: Technician = logged-in user	Closed
ISS-04	Reminder e-mail was sent for paused schedules	Schedule criteria had no status check	Added Schedule Status = Active to the W7 criteria	Closed

6.4 User Acceptance Testing

Three people used the application in an assigned role for two days and gave feedback, which is summarised in Table 6.3.

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
Priya Dharshini	Requester	The request form is simple; wanted to see where the request is waiting.	Added the Approval Status badge to the Purchase Requests report.
Murugan R.	Technician	Mobile job entry is easy; needed the option to attach a photo of the repair.	Added a Job Report file-upload field.
Dr. K. Selvam	Supervisor	Dashboard is clear; wants approvals from the phone.	Enabled push-notification approval on the mobile app.

CHAPTER 7
PUBLISHING, SHARING AND SECURITY

7.1 Publishing the Application

The application was developed in the Development environment and deployed to the Production environment on 20 October 2026. Users were invited by e-mail under Settings → Share and each was given one of the four roles. The application can be opened on the web from its URL and on the Zoho Creator mobile app after login. Public permalinks, embedding in websites and the customer portal were left disabled because the data is meant only for department staff. The publishing settings are shown in Figure 7.1, and the checklist followed is in Table 7.1.

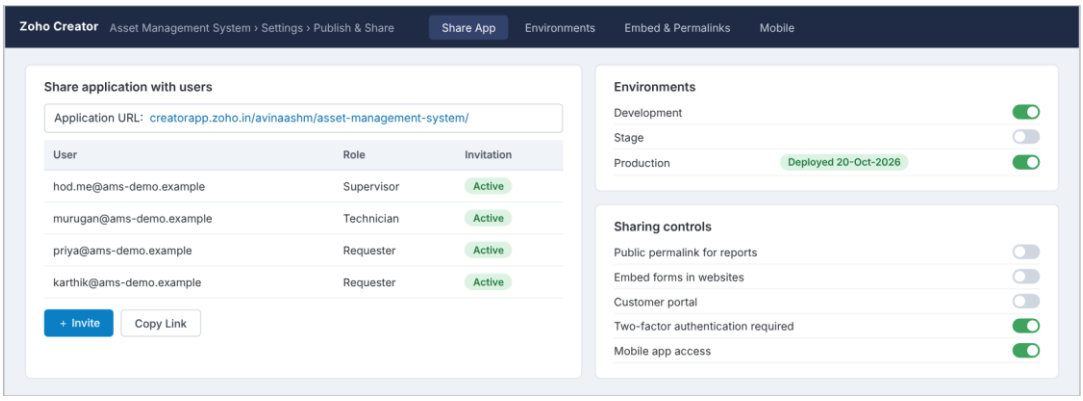


Figure 7.1: Publish and Share Settings

Table 7.1: Publishing Checklist

S.No.	Checklist Item	Done (✓)
1	All test cases executed and passed	✓
2	Test records removed	✓
3	Users invited and roles assigned	✓
4	Workflows and schedules switched on	✓
5	Application opened and checked on the mobile app	✓
6	Application link verified by another user	✓

7.2 Security and Data Protection

Security was handled with built-in settings only. Role permissions limit what each user can open, record-level rules limit users to their own or assigned records, sensitive cost fields are hidden from requesters, public links are disabled, and two-factor

authentication is required at login. The settings and the risks they prevent are in Table 7.2.

Table 7.2: Security Settings Applied

Security Setting	Where Applied	Risk Prevented
Record-level rule: own records only	Asset Request, Asset Disposal (Requester)	Users seeing others' requests
Record-level rule: assigned records only	Maintenance Record and Schedule (Technician)	Technicians changing other jobs
Field hidden for Requester role	Estimated Unit Cost and Total Cost fields	Exposure of cost details
Dashboard page restricted	Asset Dashboard (Administrator, Supervisor)	Unauthorised view of asset values
Published link disabled	Report permalinks, embedding, portal	Public access to asset data
Two-factor authentication	All users	Account misuse after password theft

7.3 Administration and Maintenance

The Administrator (asset manager) is the owner of the application. Changes such as a new category, a new location or a changed approval limit are made in the builder in the Development environment, tested, and then deployed to Production. Data can be exported from any report to Excel or CSV for backup, and Zoho also keeps automatic backups of the account. The record history (audit trail) of each record shows who added or changed it and when, which is used to settle questions about approvals and edits. Users who leave the department are deactivated, and their records remain in the application.

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The working application is presented through one realistic scenario that passes through each role. Priya Dharshini, a faculty member, needs two digital torque wrenches for the Metrology Lab. The scenario follows the request from submission to approval, purchase, allocation, a later repair job and the updated dashboard.

Step 1 – Request raised. The Requester fills the Asset Request form. The total cost of ₹37,000 is calculated automatically and, on submission, the request is sent for approval (Figure 8.1).

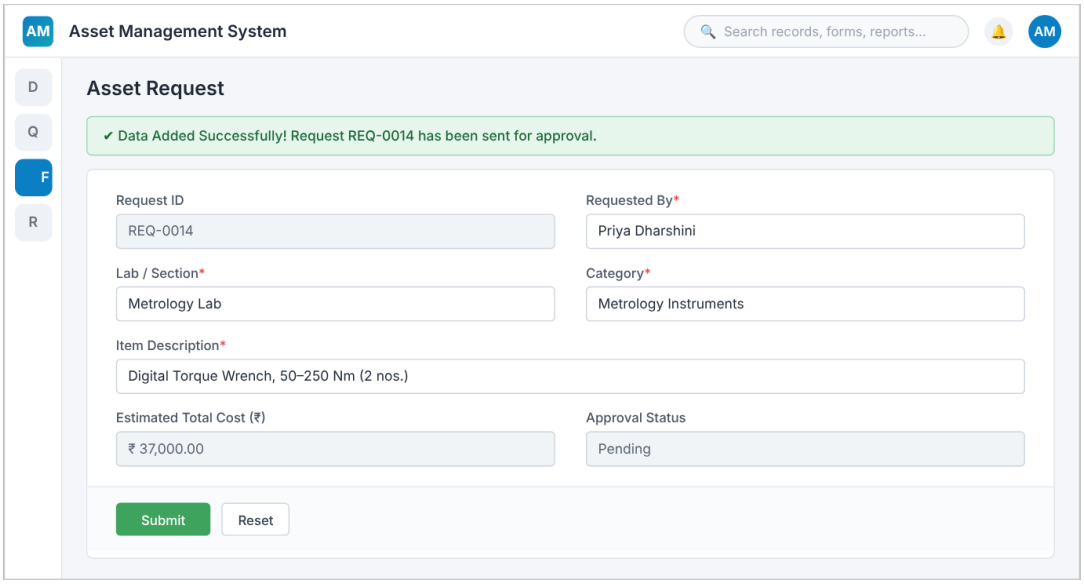


Figure 8.1: Scenario Step 1 – Requester Raises a Purchase Request

Step 2 – Request approved. The Administrator approves at Level 1 and, because the cost is above ₹25,000, the Supervisor approves at Level 2 from the My Approvals screen. The requester receives an e-mail (Figure 8.2).

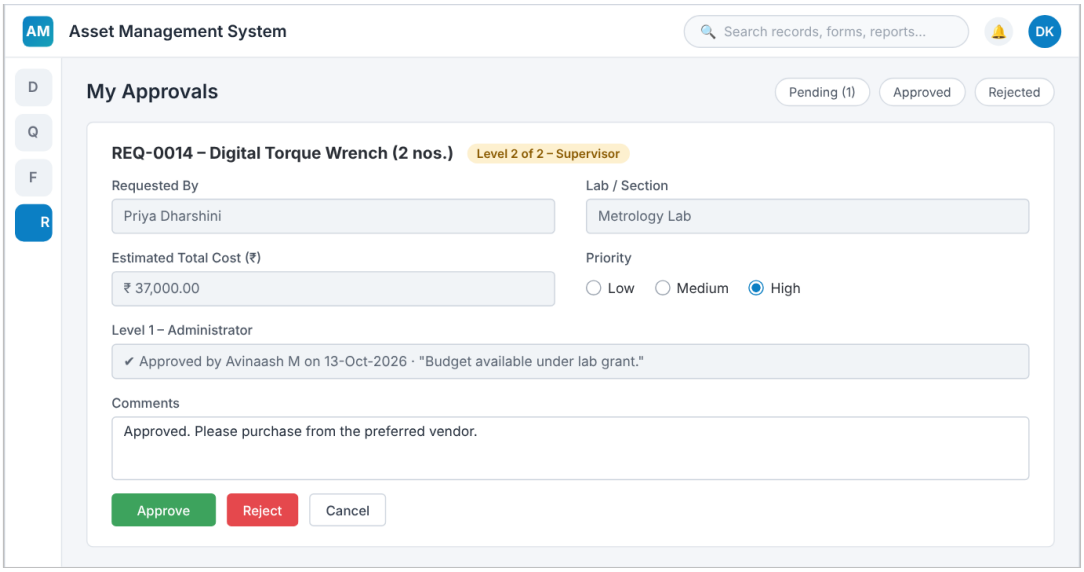


Figure 8.2: Scenario Step 2 – Supervisor Approves at Level 2

Step 3 – Assets registered and allocated. After purchase, the Administrator registers two assets (AST-0011 and AST-0012). One is allocated to the Metrology Lab, and workflow W2 changes its status to In Use while the other remains In Stock (Figure 8.3).



Figure 8.3: Scenario Step 3 – Assets Registered and Allocated

Step 4 – Maintenance job above the limit. Later the Universal Testing Machine breaks down. The technician records JOB-0027 with two spare parts and labour; the total of ₹12,800 exceeds ₹10,000 and workflow W5 sends the job to the Supervisor (Figure 8.4).

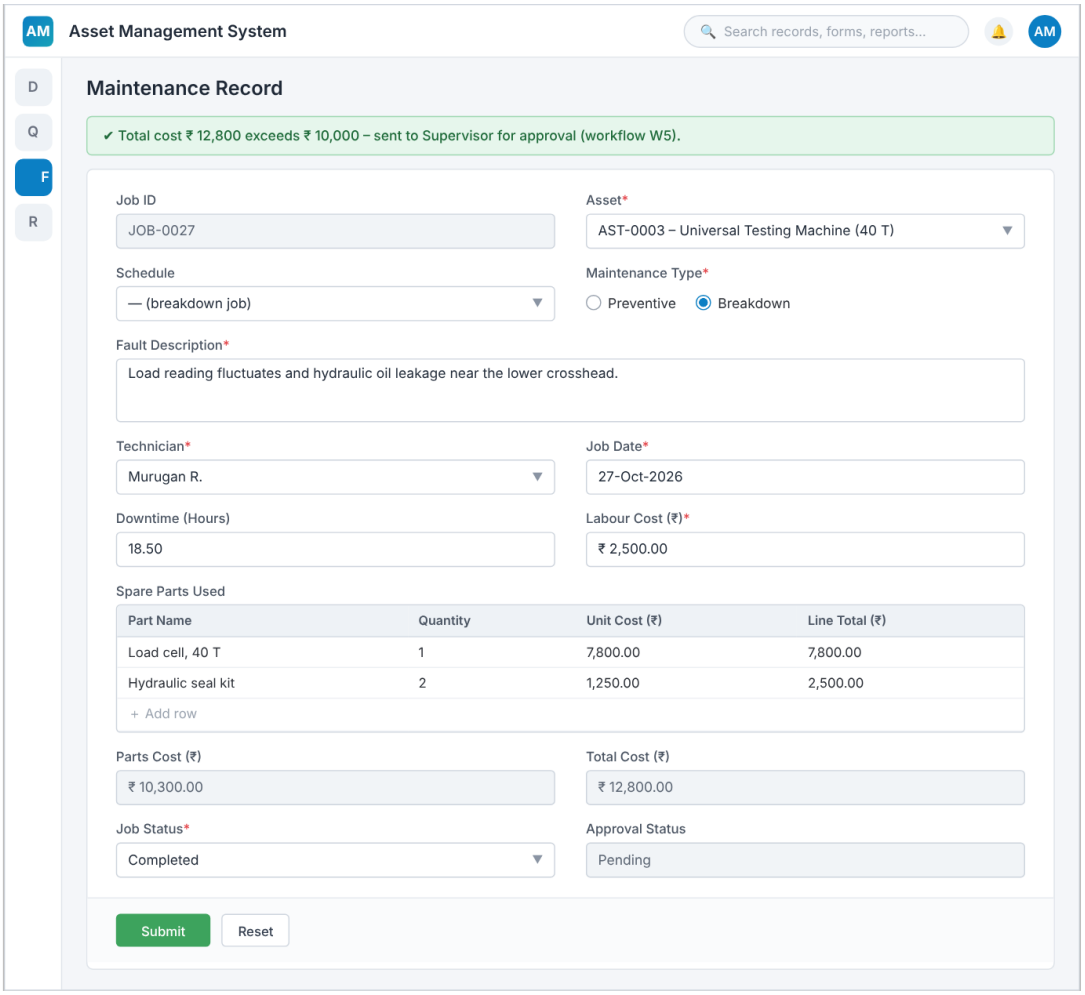


Figure 8.4: Scenario Step 4 – Technician Records a Breakdown Job

Step 5 – Dashboard updated. The dashboard now shows 150 assets, one more asset in use and one fewer pending approval, without any manual summary (Figure 8.5).

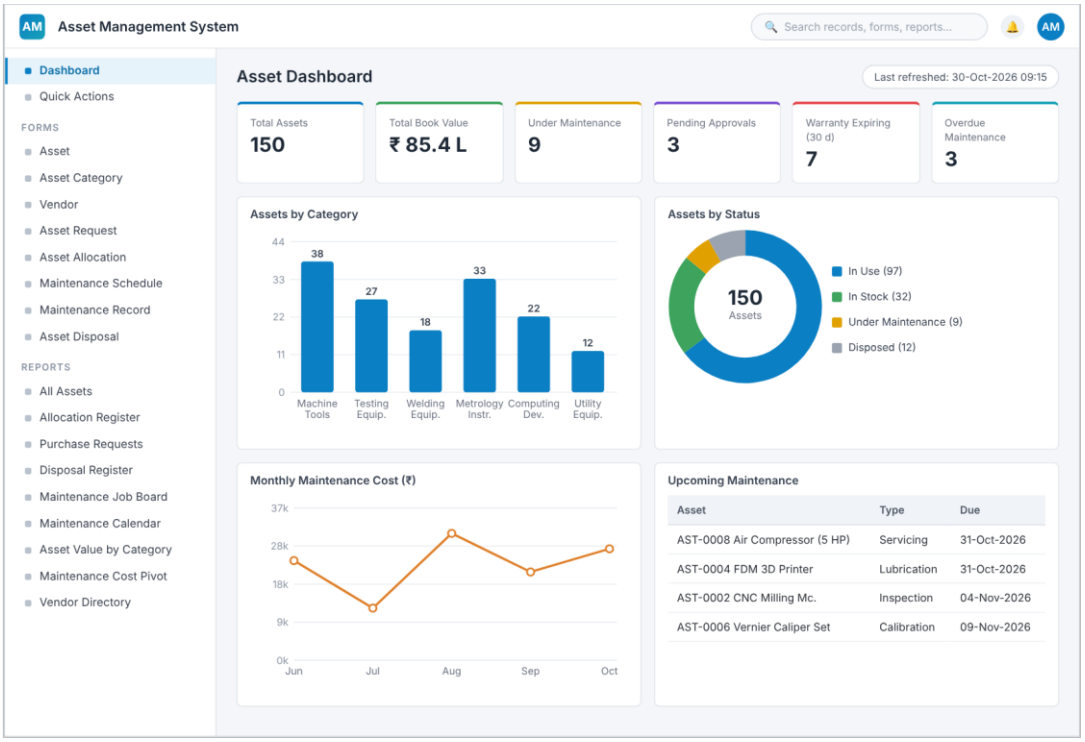


Figure 8.5: Scenario Step 5 – Dashboard After the Scenario

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Achieved Through	Outcome / Evidence
1. Digitise records with eight forms	Sections 4.3, 5.2	Eight forms with 23 field types, lookups and a subform; 148 assets imported.
2. Provide reports and a dashboard	Sections 5.4, 5.5	Nine reports of six types and two pages; KPIs and charts update automatically.
3. Automate notifications and updates	Section 5.6 (W1–W3, W7, W8)	Status updates and reminders happen without manual action.
4. Route approvals	Section 5.6 (W4–W6)	Purchase, repair-cost and disposal decisions follow set levels and conditions.
5. Restrict access by role	Sections 4.6, 5.7, 7.2	Four roles; record-level rules verified in TC-18 and ISS-03.
6. Publish for web and mobile	Sections 5.10, 7.1	Deployed on 20 October 2026; used on the mobile app.

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Time to complete one purchase request	About 5 working days	Under 1 day	About 80 % faster
Paper / manual steps	8 paper steps per asset life cycle	None for entry and approval	Paperless

Parameter	Existing System	Proposed Application	Improvement
Time to prepare summary report for the HoD	About 2 hours	Instant on the dashboard	From hours to seconds
Missed / delayed maintenance and warranty dates	Frequent, no reminders	Automatic reminders by e-mail and push	Missed dates avoided
Time to find an asset's status and location	10–15 minutes of searching	Under 1 minute with search or QR scan	About 90 % faster
Cost history per machine	Not available	Available in the Maintenance Cost report	New capability

The values for the existing system were estimated from discussions with the workshop staff, and the values for the proposed application were observed during the trial use.

8.4 Limitations of the No-Code Approach

Some needs could not be met with no-code tools alone. The depreciation formula is simple straight-line depreciation; a declining-balance method or tax-based depreciation would need scripting. Screens follow the platform's layouts, so custom screen designs and special print formats are limited. The application cannot read data from machines or sensors directly, so machine running hours are entered manually. Integration with the college's accounts software or purchase-order system would require an API connection. A low-code approach with Deluge scripting, or a custom integration, could solve these points if the department needs them in future.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Asset Management System replaces handwritten registers, issue slips and spreadsheets in a mechanical engineering workshop with a single Zoho Creator application. It has eight forms, nine reports, two pages and eight workflows of three types (three form workflows, three approval processes and two schedules), with four user roles and mobile access. The application was built without writing code, tested with twenty test cases and three acceptance users, and published on 20 October 2026.

Purchase requests, repair costs and disposals now follow clear approval levels, assets change status automatically when they are allocated, maintained or returned, and reminders reach technicians before a machine is due for service. The HoD gets the department's asset position from a dashboard instead of a manual report. The project met all six objectives set in Section 1.3.

9.2 Future Enhancements

- Print a QR code label for every asset and open its service history by scanning it with a phone.
- Use Zia predictions on maintenance history to suggest the best service interval for each machine.
- Capture running hours from machine sensors through low-code integration to trigger usage-based maintenance.
- Add a Blueprint for the repair job (Reported → Assigned → In Progress → Completed → Verified) to enforce stages.
- Connect the application to the college's accounts system for budget checks and purchase orders.

9.3 Relevance to My Discipline

For a mechanical engineer, maintenance, asset records and costs are everyday concerns, and this project showed that such processes can be digitised without depending on a software team. The skills gained include designing a data model, choosing field types, writing formulas and validation rules, building approval workflows, configuring roles and testing an application. These skills prepare a student for roles such as citizen

developer, maintenance planning engineer or digital-transformation engineer in manufacturing and service industries.

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APPENDIX A – ZOHO CREATOR NO-CODE FEATURE CHECKLIST

Each feature is marked as configured (Y) or not (N), with its location in the application and the section or figure that shows it.

Table A.1: No-Code Feature Coverage Checklist

Area	Zoho Creator Feature	Done	Unit	Where in the Application	Section / Figure
Setup	Application created; sections/menu, theme and icon configured	Y	I	Asset Management System (menu, theme, icon)	5.1; Figure 5.1
Forms	Forms using ≥ 10 distinct field types	Y	II	All 8 forms (23 field types)	4.3; Figure 5.2
Forms	Lookup field linking two forms	Y	II	Asset → Asset Category and Vendor	4.2; Figure 4.2
Forms	Subform (multiple rows in one record)	Y	II	Maintenance Record → Spare Parts Used	5.2; Figure 5.9
Forms	Mandatory, unique and default-value properties	Y	II	Asset (unique Serial Number) and all forms	4.3; Table 4.4
Forms	Media fields (Image / File / Signature / Barcode-QR)	Y	II	Asset, Asset Allocation, Asset Request	5.2; Figure 5.3
Logic	Formula field	Y	III	FM-01 to FM-08	5.3; Table 5.1
Logic	Field rules / conditional show–hide	Y	III	RL-01 to RL-06	5.3; Figure 5.11
Logic	Validation rule with error message	Y	III	VL-01 to VL-04	5.3; Figure 5.11
Reports	List report with filter, grouping and sorting	Y	III	All Assets; Purchase Requests	5.4; Figure 5.12
Reports	Kanban / Calendar / Timeline / Map report	Y	III	Maintenance Job Board; Maintenance Calendar	5.4; Figure 5.14
Reports	Pivot table or chart report	Y	III	Asset Value by Category; Cost Pivot	5.4; Figure 5.16
Reports	Conditional formatting or record template	Y	III	All Assets; Purchase Requests	5.4; Figure 5.13
Pages	Dashboard page with KPI widgets and charts	Y	IV	Asset Dashboard	5.5; Figure 5.18
Pages	Page with embedded form / report and buttons	Y	IV	Quick Actions	5.5; Figure 5.19
Workflow	Form workflow with email / notification action	Y	IV	W1, W3	5.6; Figure 5.20
Workflow	Form workflow updating a field or creating a record	Y	IV	W2, W3	5.6; Figure 5.21
Workflow	Scheduled / reminder workflow	Y	IV	W7, W8	5.6; Figure 5.27
Workflow	Approval process	Y	IV	W4, W5, W6	5.6; Figure 5.24
Workflow	Blueprint (stage-based process)	N	IV	Not used: Form, Approval and Schedule workflows were requested	5.6
Users	Users invited; roles and permissions configured	Y	V	4 roles, 6 users	5.7; Figure 5.29
Users	Record-level access rule	Y	V	Technician and Requester rules	5.7; Table 7.2

Area	Zoho Creator Feature	Done	Unit	Where in the Application	Section / Figure
Data	Import from Excel / CSV	Y	V	Asset form import	5.8; Figure 5.30
Data	Export / share report (PDF, Excel, link)	Y	V	Report export menu	5.8
Data	Built-in integration with another Zoho app	Y	V	Zoho Mail, Zoho Sheet	5.8
AI	AI app builder / Zia feature used	Y	V	Zia AI application builder	5.9; Figure 5.31
Publish	Application published to users / portal	Y	V	Production environment, 20 October 2026	7.1; Figure 7.1
Publish	Used on the Zoho Creator mobile app	Y	V	Zoho Creator mobile app	5.10; Figure 5.32
Security	Security settings applied (hidden fields, link control, 2FA)	Y	V	Hidden cost fields, links off, 2FA	7.2; Table 7.2
Testing	Test cases and user acceptance testing documented	Y	V	20 test cases; 3 UAT users	6.2, 6.4

Total features configured: 29 / 30

APPENDIX B – COURSE OUTCOME MAPPING

The course outcomes are mapped to the chapters of this report, with a self-assessed attainment level (1 – Low, 2 – Medium, 3 – High).

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the concepts of no-code and low-code development, citizen development and the architecture of the Zoho Creator platform.	Ch. 1, 2	3	
CO2	Design and build forms with suitable field types, validations, formulas and rules for a real engineering problem.	Ch. 3, 4, 5.2, 5.3	3	
CO3	Create reports, pages and dashboards that present data to different users.	Ch. 5.4, 5.5	3	
CO4	Automate business processes using form workflows, schedules and approval processes.	Ch. 5.6	3	
CO5	Configure users, roles, security, publishing and mobile access, and test the application.	Ch. 5.7–5.10, 6, 7	3	

APPENDIX C – APPLICATION CONFIGURATION RECORD

This record lists every component configured, so that another person can maintain or rebuild the application.

Table C.1: Summary of Application Components

Component Type	Name	Purpose	Section Ref.
Form	Asset Category	Master list of asset categories with default depreciation rate	4.3, 5.2
Form	Vendor	Master list of suppliers and service providers	4.3, 5.2
Form	Asset	Central register of every asset with cost, location, warranty and status	4.3, 5.2
Form	Asset Request	Purchase request raised by staff and routed through the approval process	4.3, 5.2
Form	Asset Allocation	Issue and return record of assets given to staff, labs or sections	4.3, 5.2
Form	Maintenance Schedule	Preventive maintenance plan for each asset with automatic next-due date	4.3, 5.2
Form	Maintenance Record	Job card for preventive and breakdown maintenance with spare parts and costs	4.3, 5.2
Form	Asset Disposal	Request to retire, scrap or sell an asset with two-level approval	4.3, 5.2
Report	All Assets	List report on Asset	5.4
Report	Allocation Register	List report on Asset Allocation	5.4
Report	Purchase Requests	List report on Asset Request	5.4
Report	Disposal Register	List report on Asset Disposal	5.4
Report	Maintenance Job Board	Kanban report on Maintenance Record	5.4
Report	Maintenance Calendar	Calendar report on Maintenance Schedule	5.4
Report	Asset Value by Category	Chart (Bar) report on Asset	5.4
Report	Maintenance Cost Pivot	Pivot report on Maintenance Record	5.4
Report	Vendor Directory	Spreadsheet report on Vendor	5.4
Page	Asset Dashboard	KPI tiles and charts for Administrator and Supervisor	5.5
Page	Quick Actions	Buttons, embedded report and quick-request form for all roles	5.5
Workflow	W1 – Asset Registration Alert	Form workflow on Asset	5.6
Workflow	W2 – Allocation Status Updater	Form workflow on Asset Allocation	5.6
Workflow	W3 – Maintenance Completion Updater	Form workflow on Maintenance Record	5.6
Workflow	W4 – Purchase Request Approval	Approval on Asset Request	5.6
Workflow	W5 – Repair Cost Approval	Approval on Maintenance Record	5.6
Workflow	W6 – Asset Disposal Approval	Approval on Asset Disposal	5.6

Component Type	Name	Purpose	Section Ref.
Workflow	W7 – Preventive Maintenance Reminder	Schedule on Maintenance Schedule	5.6
Workflow	W8 – Warranty Expiry Alert	Schedule on Asset	5.6
Role	Administrator	Asset manager; full access to forms and reports	4.6, 5.7
Role	Supervisor	HoD; approvals, dashboard and reports	4.6, 5.7
Role	Technician	Assigned schedules and own maintenance records	4.6, 5.7
Role	Requester	Own requests and disposals	4.6, 5.7

Table C.2: Workflow Settings in Detail – W2 Allocation Status Updater

Form / Report	Asset Allocation
Run when	Record created or edited → On success
Condition	If Allocation Status = Issued; else if Actual Return Date is not empty
Action 1	Update records (Asset) → Criteria: Asset ID = input.Asset; Status = In Use
Action 2	Update records (Asset) → Status = In Stock
Action 3	Update this record → Allocation Status = Returned

Table C.3: Workflow Settings in Detail – W4 Purchase Request Approval

Form / Report	Asset Request
Run when	Record created
Condition	Level 2 applies when Estimated Total Cost > 25,000
Action 1	Level 1 approval → Administrator; reminder after 2 days
Action 2	Level 2 approval → Supervisor
Action 3	On approve: Approval Status = Approved; send e-mail to Requested By. On reject: Approval Status = Rejected; send e-mail with comments

Table C.4: Workflow Settings in Detail – W7 Preventive Maintenance Reminder

Form / Report	Maintenance Schedule
Run when	Scheduled: daily at 08:00 AM, starting 20 October 2026
Condition	Next Due Date ≤ today + 3 days and Schedule Status = Active
Action 1	Send e-mail → To: Assigned Technician | Subject: Maintenance due – asset name
Action 2	Send push notification → Assigned Technician
Action 3	If Next Due Date < today: send e-mail → Supervisor | Subject: Overdue maintenance

APPENDIX D – USER MANUAL

D.1 Accessing the Application

On a computer, open https://creatorapp.zoho.in/avinaashm/asset-management-system/ in Chrome, Edge or Firefox and sign in with the e-mail address and password given by the Administrator; enter the verification code when asked. On a phone, install the Zoho Creator app from the Play Store or App Store, sign in with the same account and open Asset Management System from the application list. The menu shows only the items allowed for your role.

D.2 Instructions for the Administrator

- Add categories and vendors from the Asset Category and Vendor forms before registering assets.
- Register each asset in the Asset form; scan the tag with the mobile app and attach a photograph.
- Approve purchase requests and disposals from My Approvals; open the dashboard for the overall position.
- Issue an asset from the Asset Allocation form and enter the actual return date when it comes back.
- Create a Maintenance Schedule for each machine and assign a technician.

D.3 Instructions for the Supervisor

- Open the Asset Dashboard to see pending approvals, warranty alerts and overdue maintenance.
- Approve or reject purchase requests above ₹25,000, repair costs above ₹10,000 and disposals, adding comments.
- Use the Purchase Requests, Disposal Register and Maintenance Cost Pivot reports for audits.

D.4 Instructions for the Technician

- Check the Maintenance Calendar and the e-mail or push reminder for work due.
- Open the Maintenance Record form, select the asset and schedule, and enter the work done.

- Add each spare part in the subform, enter the labour cost and set the Job Status to Completed.
- Track jobs on the Maintenance Job Board.

D.5 Instructions for the Requester

- Open Quick Actions and choose Raise Purchase Request, or open the Asset Request form.
- Enter the item, quantity, estimated cost, priority, required-by date and justification, and press Submit.
- Follow the status in the Purchase Requests report; you will also receive an e-mail when the request is approved or rejected.
- Use the Asset Disposal form to ask for an obsolete asset to be retired.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	AVINAASH M	**Register No.**	241101014
Branch	Mechanical Engineering	**Date**	31 October 2026

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Relevance to own discipline, clear objectives, complete FR/NFR, use cases	10	
2	Application Design	Data model, form specifications, process flows, permission matrix	10	
3	Forms, Formula Fields & Rules	Variety of field types, lookups/subforms, formulas, field and validation rules	15	
4	Reports, Pages & Dashboards	Variety of report types, filters/formatting, dashboard design	15	
5	Workflows & Automation	Form workflows, schedules, approvals, Blueprint configured without code	15	
6	Users, Security, Data & Publishing	Roles and permissions, record rules, import/export, publishing, mobile app	10	
7	Live Demonstration	End-to-end working across roles; feature checklist verified	10	
8	Project Report	Adherence to template and format, clarity, screenshots, references	5	
9	Viva-Voce	Understanding of the platform, design choices and no-code concepts	10	
	TOTAL		**100**	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature configured completely and correctly, beyond the basic requirement; student explains choices confidently.
Good	70–89	Feature configured correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially configured or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, not working, or not understood by the student.

Signature of Internal Examiner

Name: _____

Signature of External Examiner

Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	AVINAASH M	**Register No.**	241101014
Branch / Section	Mechanical Engineering	**Date of Viva**	31 October 2026
Application URL Verified	Yes / No	**Demo Duration**	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER

Name: _____

EXTERNAL EXAMINER

Name: _____